

CAPACITANCE & CURRENT ELECTRICITY

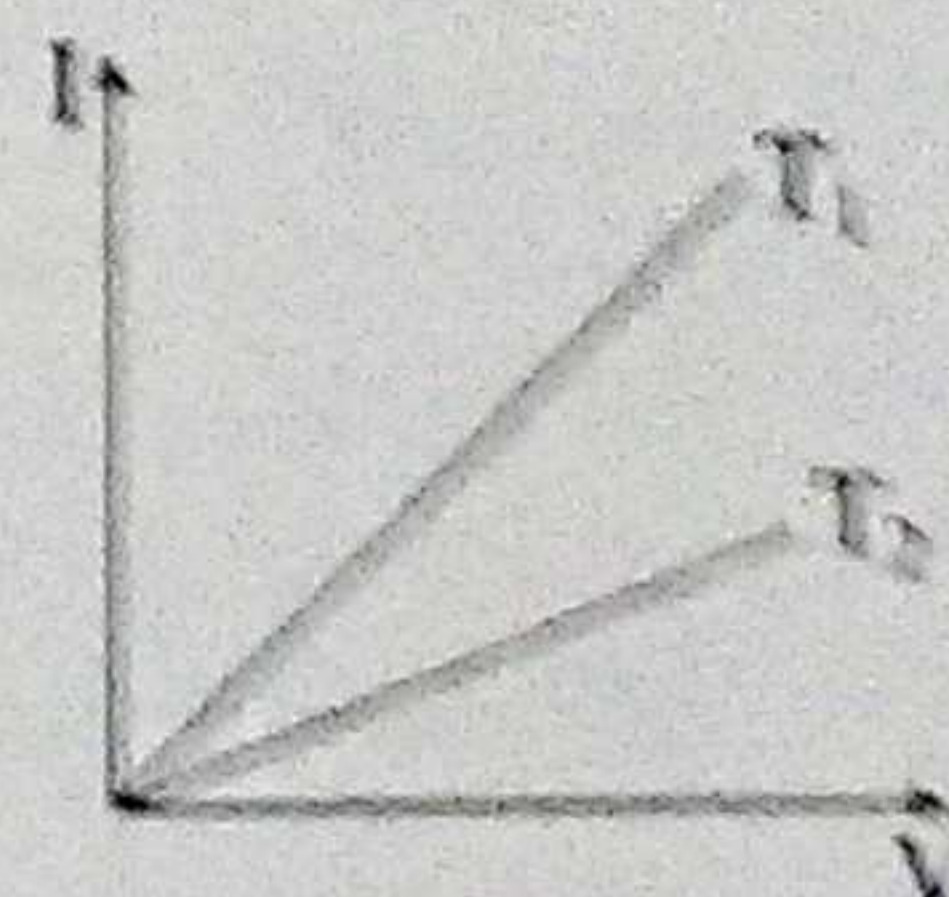
Type of CPP - Objective & Subjective

Level - MODERATE

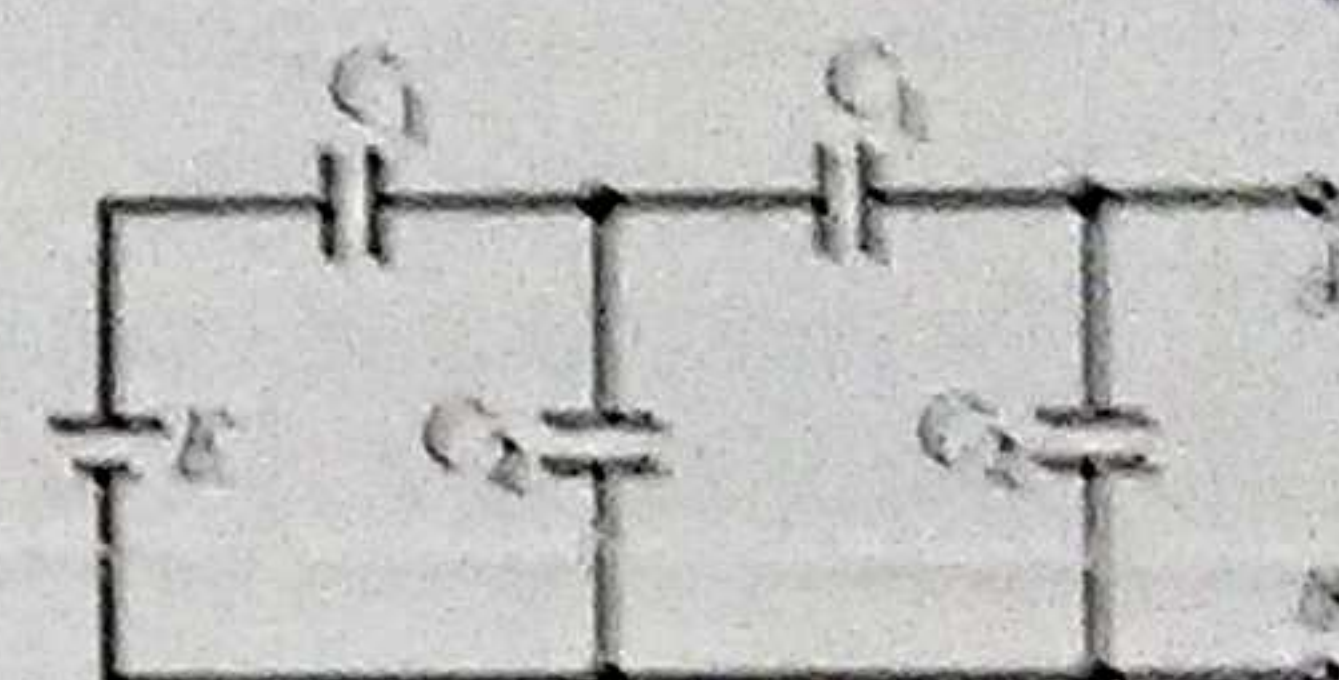
1. A current passes through a wire of nonuniform cross-section. Which of the following quantities are independent of the cross-section ?
 (A) The charge crossing in a given time interval (B) Drift speed
 (C) Current density (D) Free- electron density

2. (i) A potential difference of 200 volt is applied to a coil at a temperature of 15°C and the current is 10A. what will be the mean temperature of the coil when the current has fallen to 9 A, the applied voltage being the same as before?
 Given : temperature coefficient of resistance $(\alpha) = \frac{1}{234}^{\circ}\text{C}^{-1}$.
 (ii) A platinum wire has resistance of 10 ohm at 0°C and 20 ohm at 273°C . Find the value of temperature coefficient of resistance.

3. The current-voltage graphs for a given metallic wire at two different temperature T_1 and T_2 are shown in the figure. Which one is higher, T_1 or T_2 .



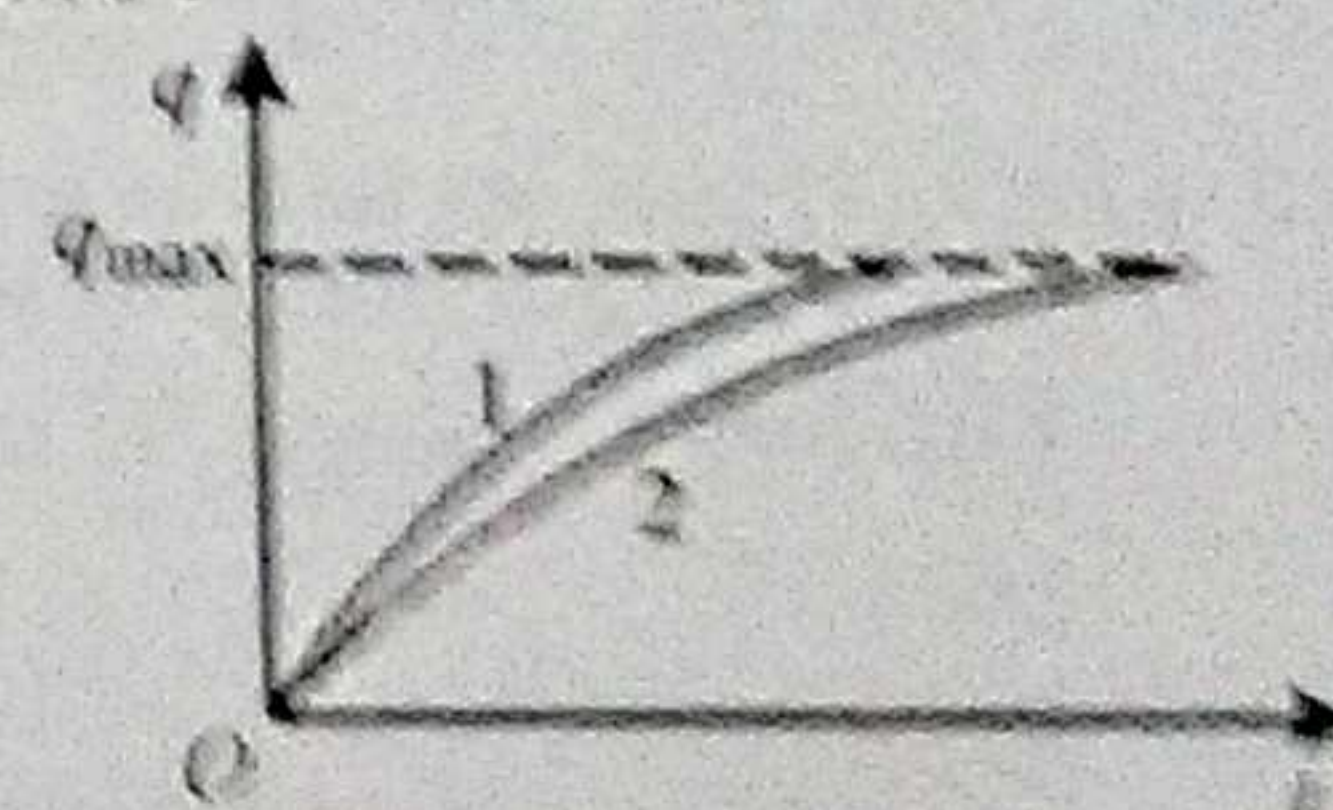
4. Find the potential difference between points A and B of the system shown in figure if the emf is equal to $E = 110\text{ V}$ and the capacitance ratio $C_2/C_1 = \eta = 2.0$.



5. The charge on capacitor in two different RC circuits 1 and 2 are plotted as shown in figure :

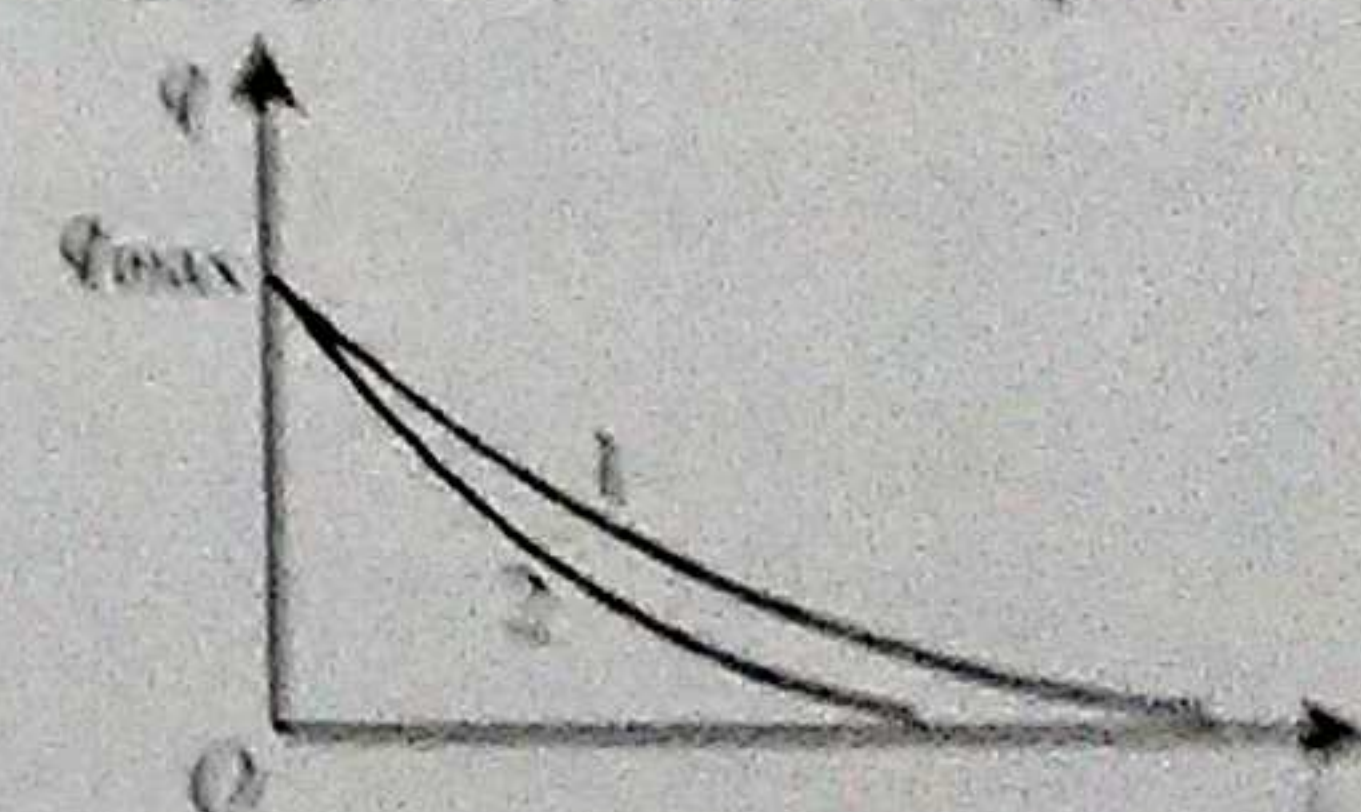
Choose the correct statement(s) related to the two circuits.

- (A) Both the capacitors are charged to the same magnitude of charge.
 (B) The emf's of cells in both the circuits are equal.
 (C) The emf's of the cells may be different.
 (D) The emf E_1 is more than E_2



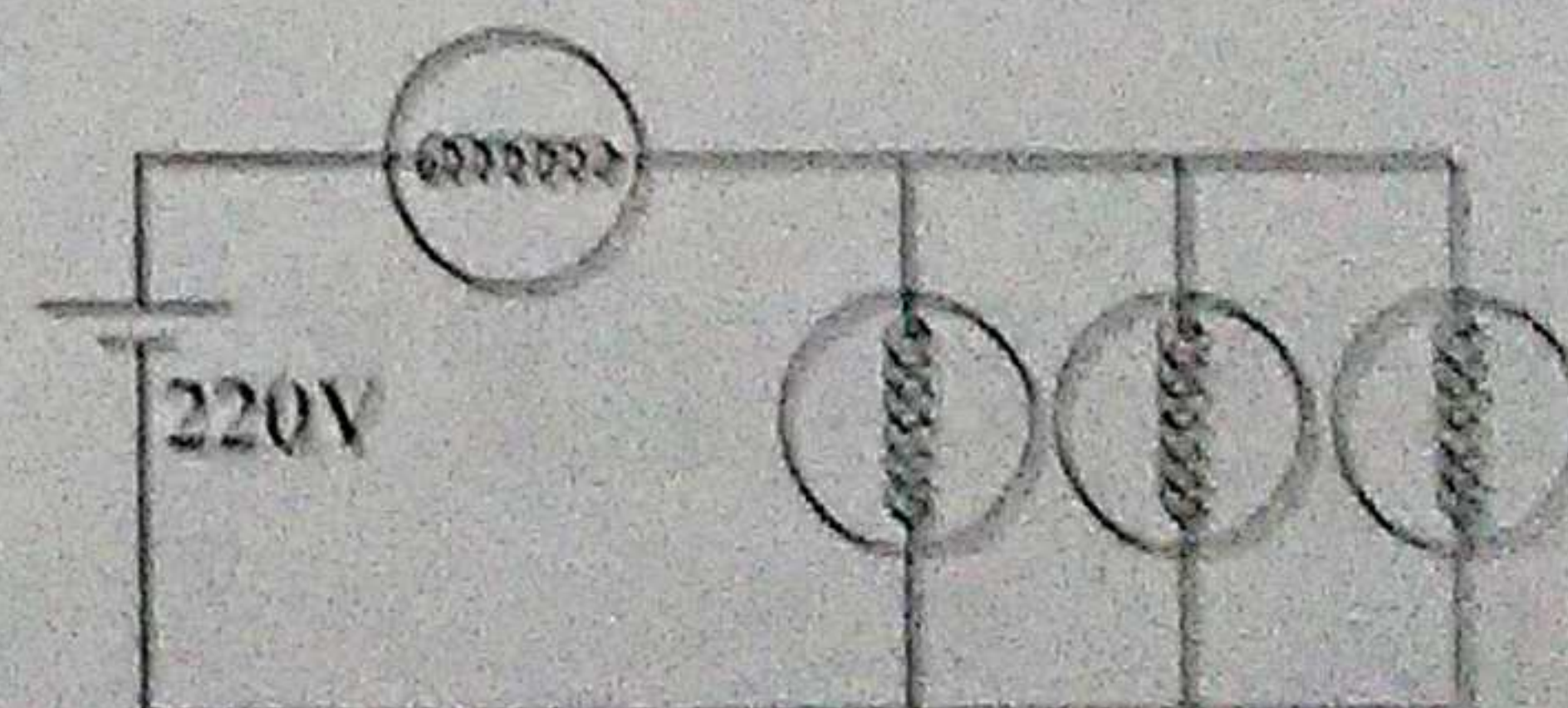
6. The instantaneous charge on capacitor in two discharging RC circuits is plotted with respect to time in figure. Choose the correct statement(s) (where E_1 and E_2 are emf's of two DC sources in two different charging circuits and capacitor are fully charged).

- (A) $R_1 C_1 > R_2 C_2$ (B) $\frac{R_1}{R_2} < \frac{C_2}{C_1}$
 (C) $R_1 > R_2$ if $E_1 = E_2$ (D) $C_2 > C_1$ if $E_1 = E_2$

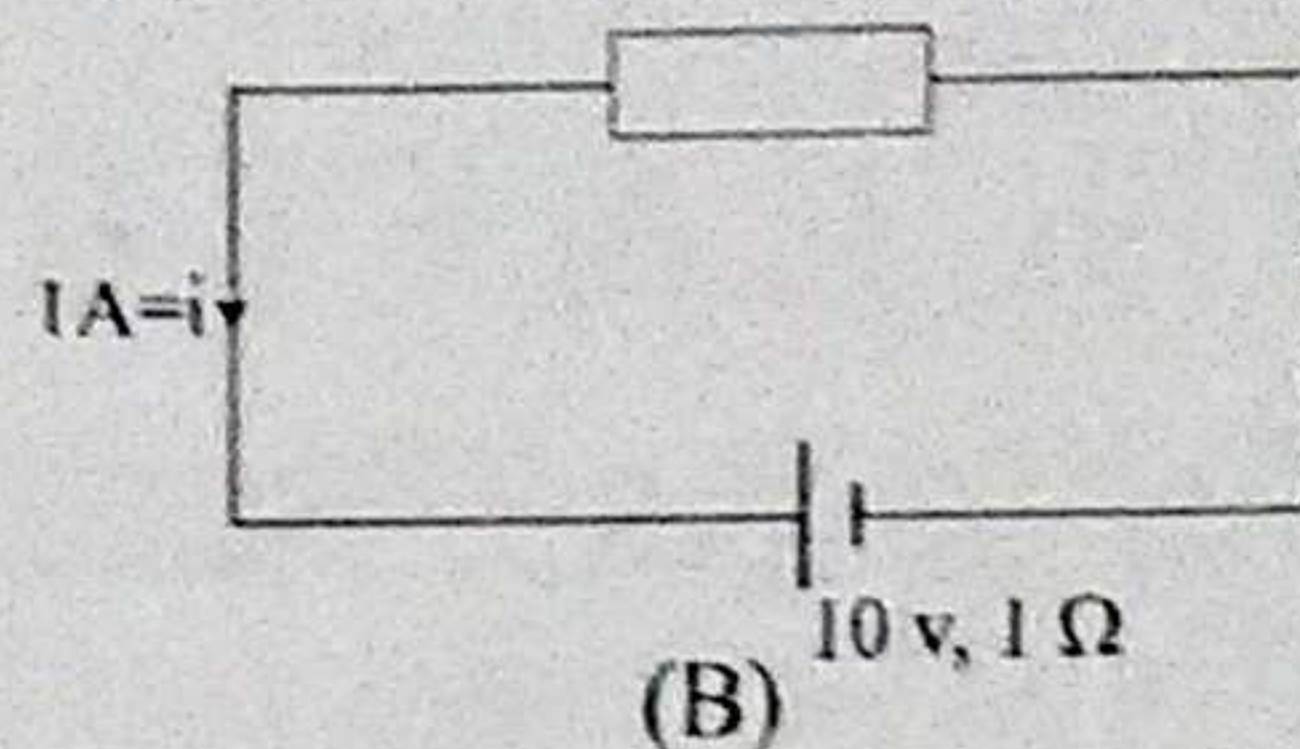
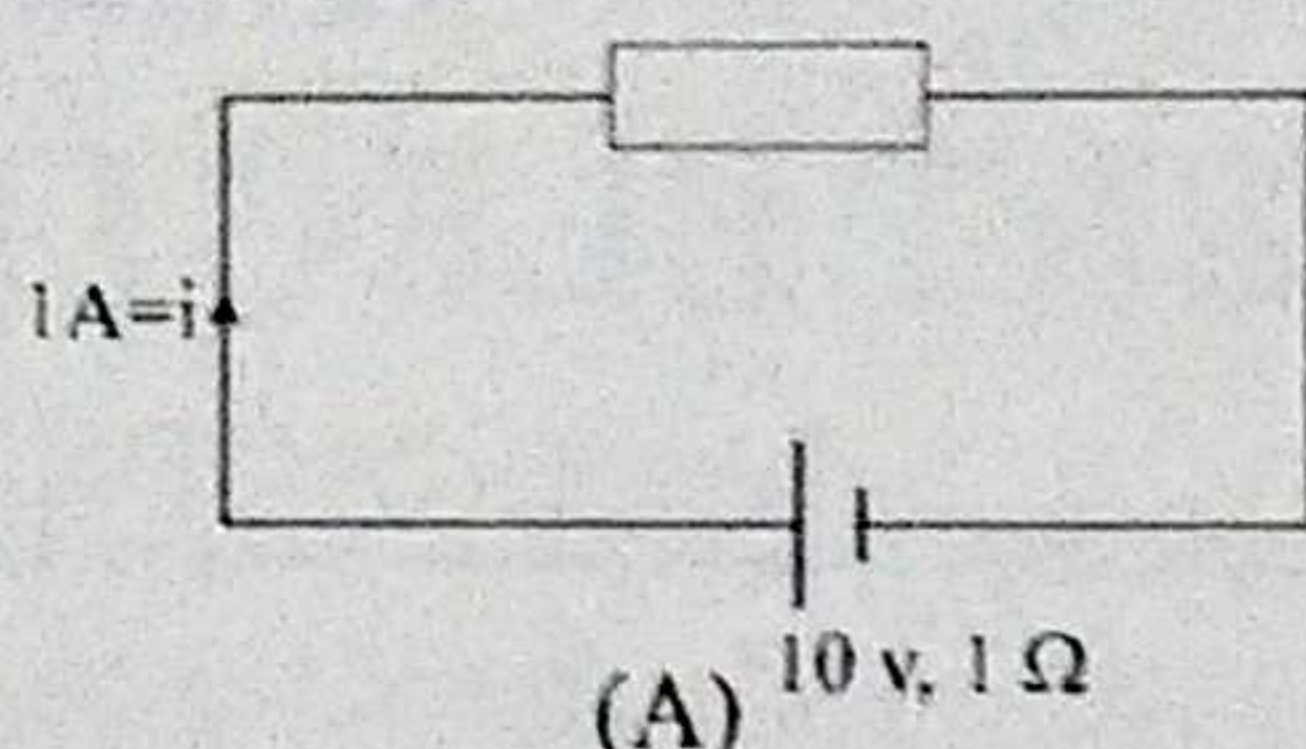


7. Four identical bulbs each rated 100 watt, 220 volts are connected across a battery as shown. The total electric power consumed by the bulbs is :

- (A) 75 watt
 (B) 400 watt
 (C) 300 watt
 (D) 400/3 watt



8. In following diagram boxes may contain resistor or battery or any other element then determine in each case :
- (a) E.m.f. of battery (b) Battery is acting as a source or load
 (c) Potential difference across each battery (d) Power input to the battery or output by the battery.
 (e) The rate at which heat is generated inside the battery.
 (f) The rate at which the chemical energy of the cell is consumed or increased.
 (g) Potential difference across box. (h) Electric power output across box.



9. The current density in a wire is 10 A/cm^2 and the electric field in the wire is 5 V/cm . If ρ = resistivity of material, σ = conductivity of the material then (in S.I. units) :
 (A) $\rho = 5 \times 10^{-3}$ (B) $\rho = 200$ (C) $\sigma = 5 \times 10^{-3}$ (D) $\sigma = 200$
10. A piece of copper and another of germanium are cooled from room temperature to 80 K . The resistance of :
 (A) Each of the them increases (B) Each of them decreases
 (C) Copper increases and germanium decreases (D) Copper decreases and germanium increases

11. The figure shows the current I in a single-loop circuit with a battery B and resistance R (and wires of negligible resistance) :

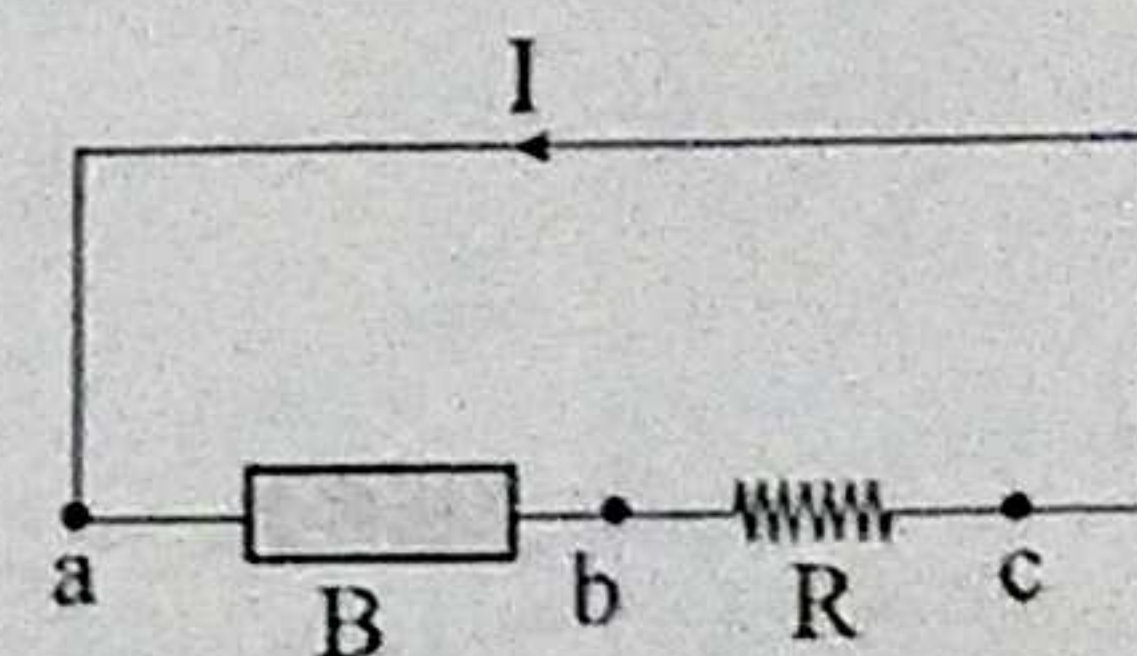
(a) Should the emf arrow at B be drawn leftward or rightward ?

At points a , b and c , rank

(b) The magnitude of the current,

(c) The electric potential, and

(d) The electric potential energy of the charge carriers (electron), greatest first



12. Two coils connected in series have resistances 600Ω and 300Ω at 20°C and temperature coefficient of resistivity 0.001 K^{-1} and 0.004 K^{-1} respectively :

- (i) The resistance of the combination at temperature 50°C is

(A) 426Ω (B) 954Ω (C) 1806Ω (D) 214Ω

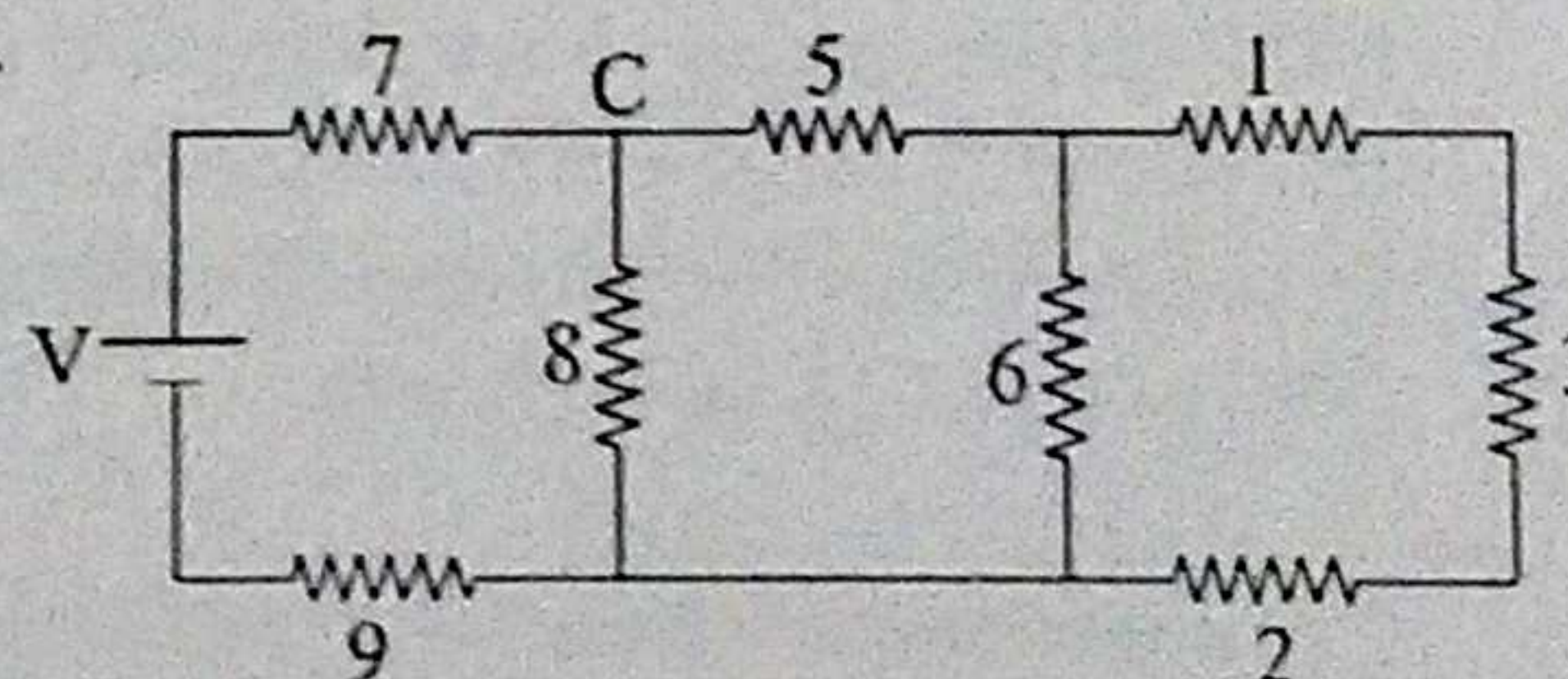
- (ii) The effective temperature coefficient of the combination is :

(A) $\frac{1}{1000} \text{ degree}^{-1}$ (B) $\frac{1}{250} \text{ degree}^{-1}$ (C) $\frac{1}{500} \text{ degree}^{-1}$ (D) $\frac{3}{1000} \text{ degree}^{-1}$

13. In the ladder network shown, current through the resistor 3Ω is 0.25 A .

The input voltage 'V' is equal to :

(A) 10 V
 (B) 20 V
 (C) 5 V
 (D) $15/2 \text{ V}$

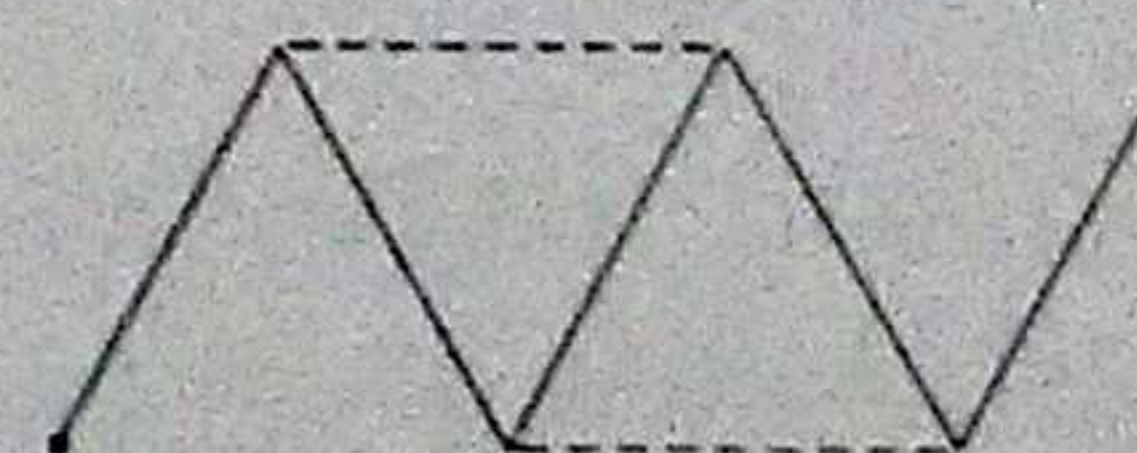


14. If 2 bulbs rated $2.5 \text{ W} - 110 \text{ V}$ and $100 \text{ W} - 110 \text{ V}$ are connected in series to a 220 V supply then :

(A) 2.5 W bulb will fuse (B) 100 W bulb will fuse (C) Both will fuse (D) Both will not fuse

15. Five identical resistor each of resistance 1Ω are initially arranged as shown in the figure by clear lines. If two similar resistances are added as shown by the dashed lines then change in resistance in final and initial arrangement is :

(A) 2Ω (B) 1Ω
 (C) 3Ω (D) 4Ω



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ANSWER KEY

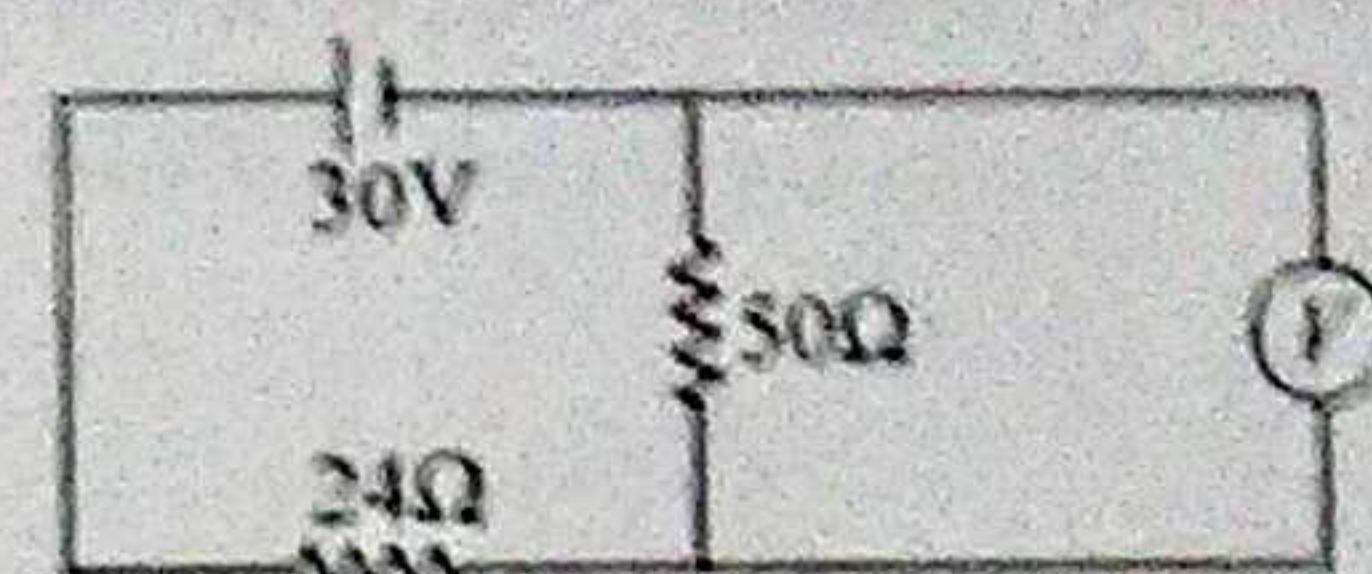
1. (A)(D) 2. [(i) 41°C ; (ii) $\frac{\ln 2}{273} ^\circ\text{C}^{-1}$] 3. $[T_2]$ 4. $[U = E/(1 + 3\eta + \eta^2) = 10 \text{ V}]$ 5. (A)(C)
 6. (A)(C) 7. (A)
 8. [(a) $E = 10 \text{ V}$ each; (b) (A) act as a source and (B) act as load; (c) $V_A = 9 \text{ V}$, $V_B = 11 \text{ V}$; (d) $P_A = 9 \text{ W}$, $P_B = 11 \text{ W}$;
 (e) Heat rate = 1 W each; (f) 10 watt , each; (g) 9 V , 11 V ; (h) -9 W , 11 W]
 9. (A)(D) 10. (D) 11. [(a) rightward; (b) all equal; (c) b , then a and c equal (d) a , c equal, b] 12. (i) (B) (ii) (C)
 13. (B) 14. (A) 15. (A)

CAPACITANCE & CURRENT ELECTRICITY

Type of CPP - Objective & Subjective

Level - MODERATE

1. The voltmeter shown in figure reads 18 V across the $50\ \Omega$ resistor. Find the resistance of the voltmeter.

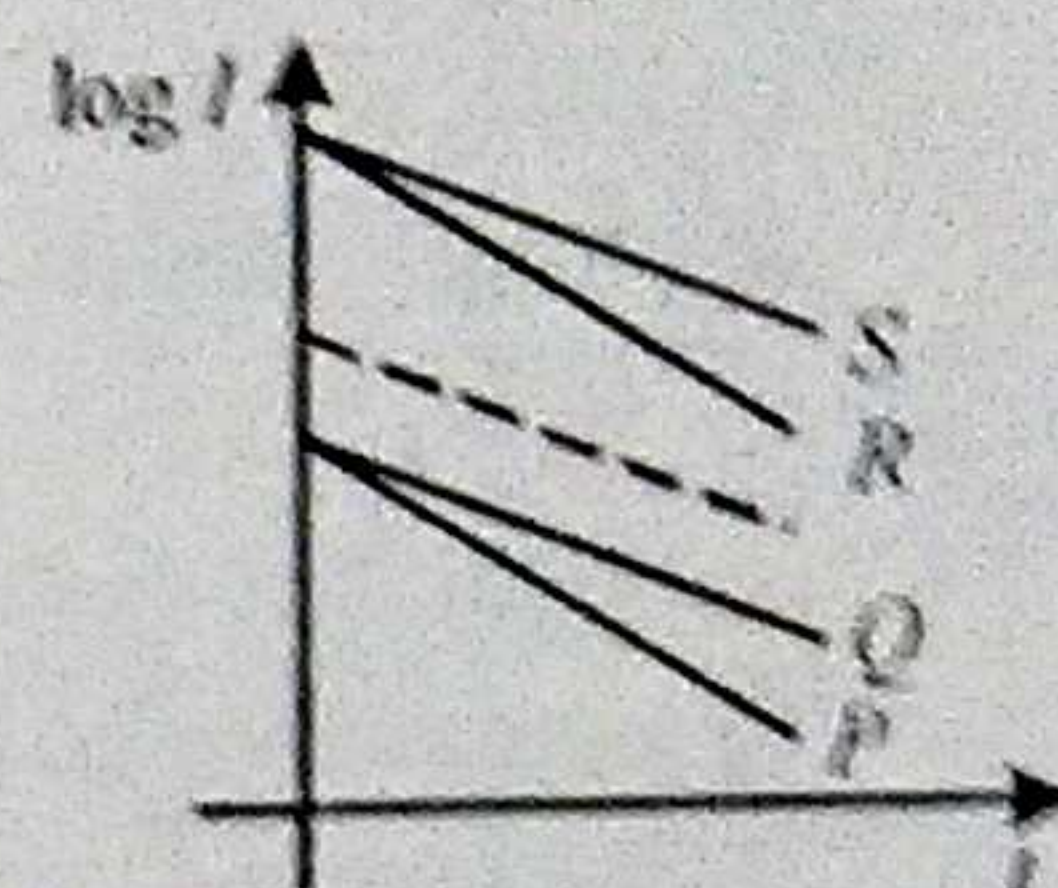


2. Two identical capacitors have the same capacitance C . One of them is charged to potential V_1 and the other to V_2 . The negative ends of the capacitors are connected together. When the positive ends are also connected, the decrease in energy of the combined system is :

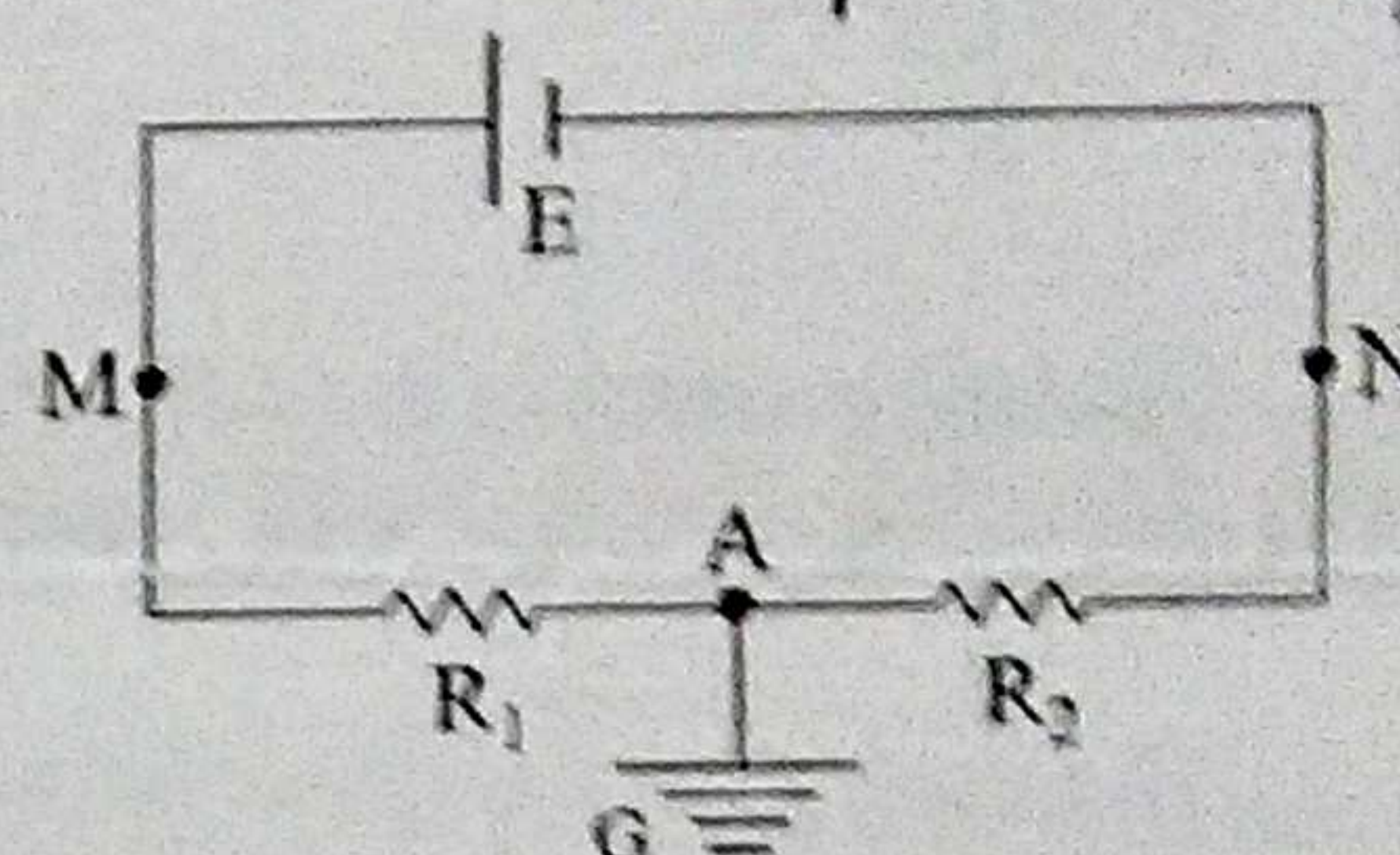
(A) $\frac{1}{4} C (V_1^2 - V_2^2)$ (B) $\frac{1}{4} C (V_1^2 + V_2^2)$ (C) $\frac{1}{4} C (V_1 - V_2)^2$ (D) $\frac{1}{4} C (V_1 + V_2)^2$

3. Dotted line represents the charging of a capacitor with resistance X . If resistance is made $2X$ then which will be the graph of charging :

- (A) P
(B) Q
(C) R
(D) S



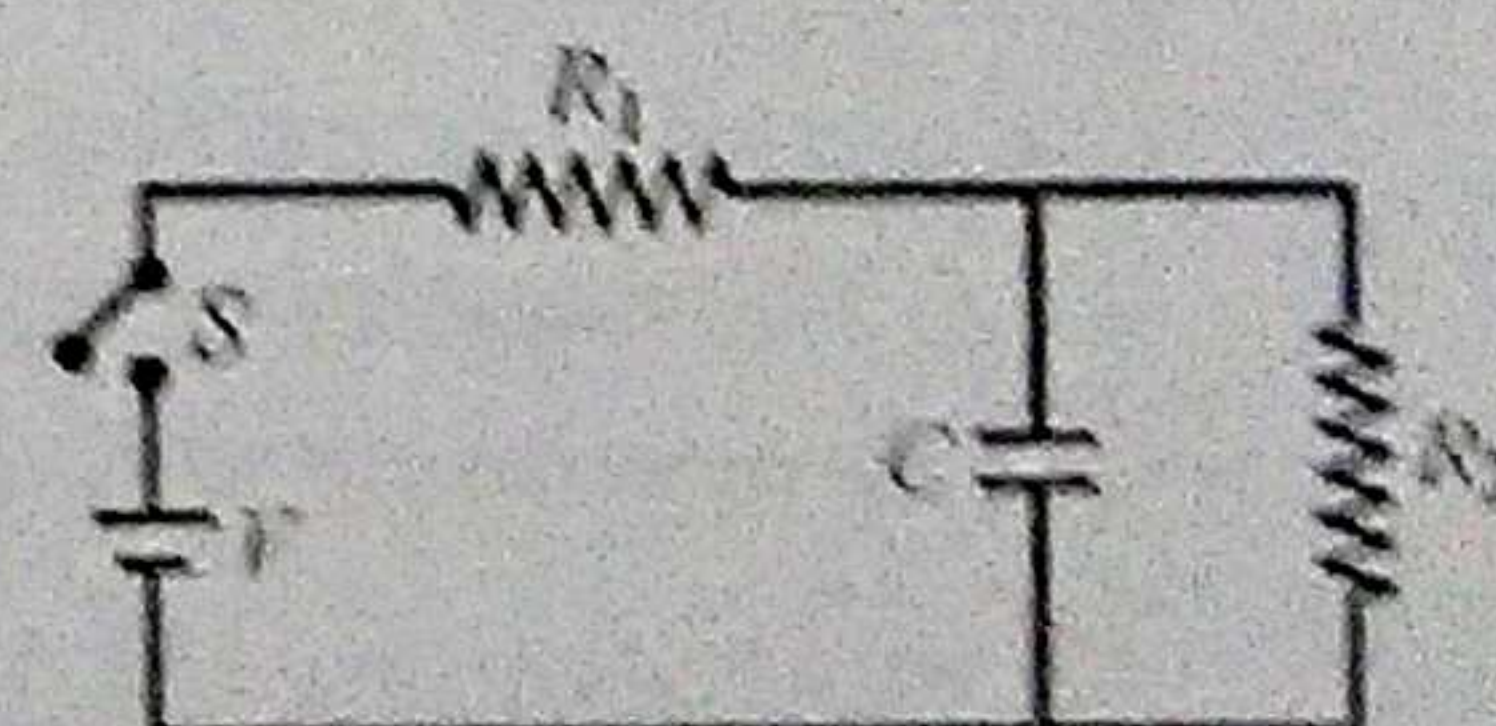
4. In the given figure, calculate the potentials at the points M and N if $E = 12\text{V}$, $R_1 = 3\ \Omega$, $R_2 = 2\ \Omega$ and $r = 1\ \Omega$. Also find current in wire AG.



5. Three bulbs, each having a resistance of $180\ \Omega$, are connected in parallel to an ideal battery of emf 60 V. Find the current delivered by the battery when (a) all the bulbs are switched on, (b) two of the bulbs are switched on and (c) Only one bulb is switched on :

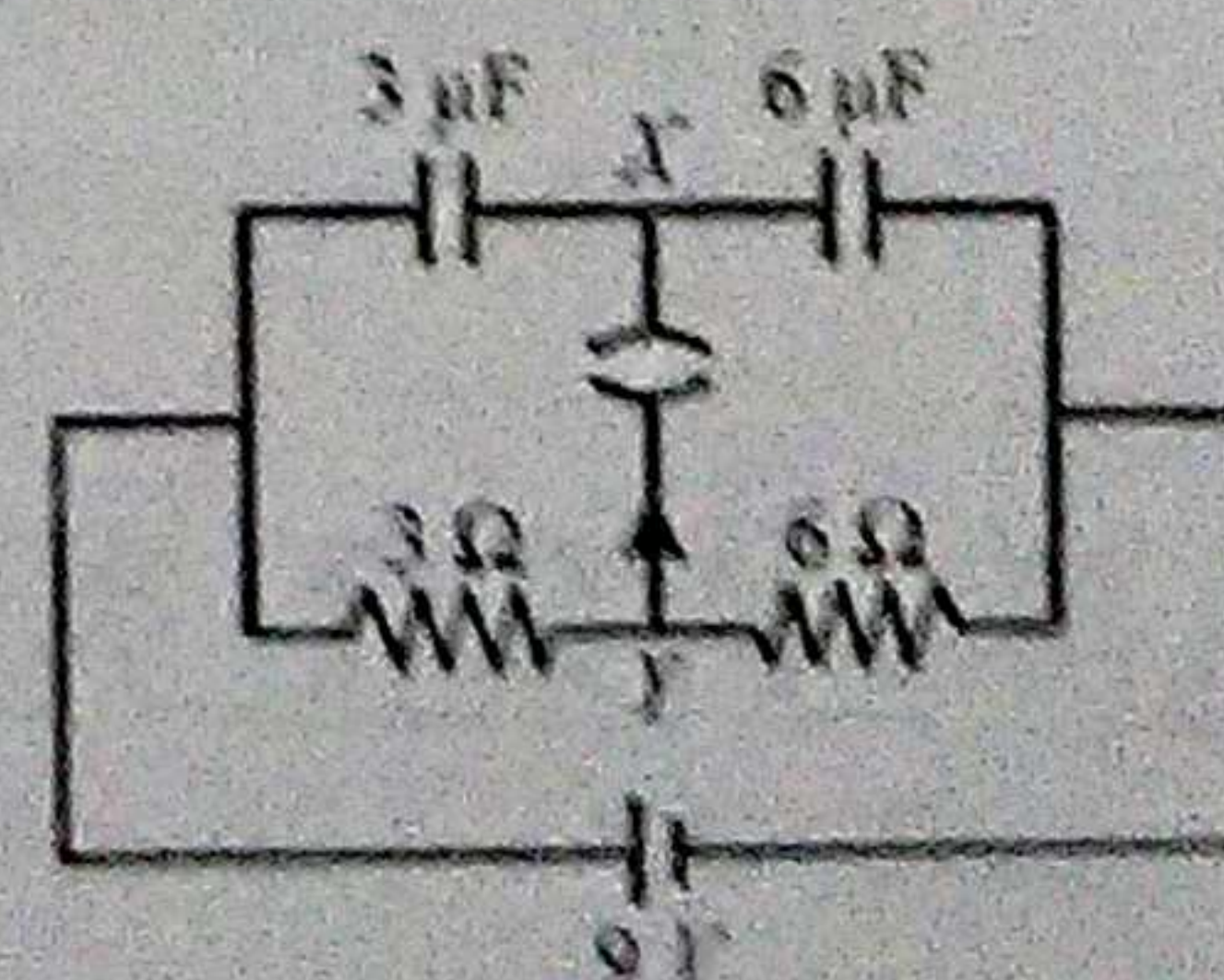
6. The resistance of a metallic conductor increases with temperature due to :
(A) Change in carrier density (B) Change in the dimensions of the conductor
(C) Increase in the number of collisions among the carriers
(D) Increase in the rate of collisions between the carriers and the vibrating atoms of the conductor

7. In the given circuit the capacitor C is uncharged initially and switch 'S' is closed at $t = 0$. If charge on capacitor at time ' t ' is given by equation $Q = Q_0 (1 - e^{-\alpha t})$. Find value of Q_0 and α .

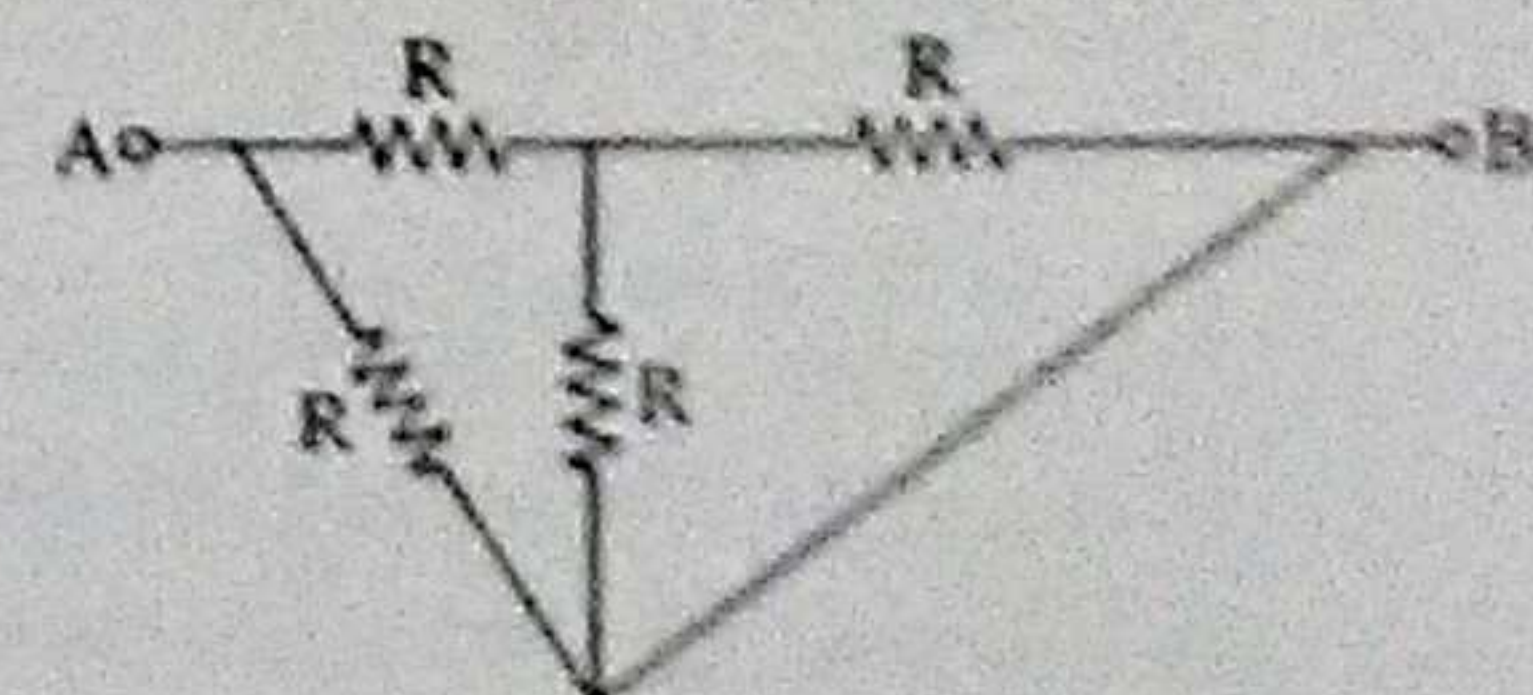
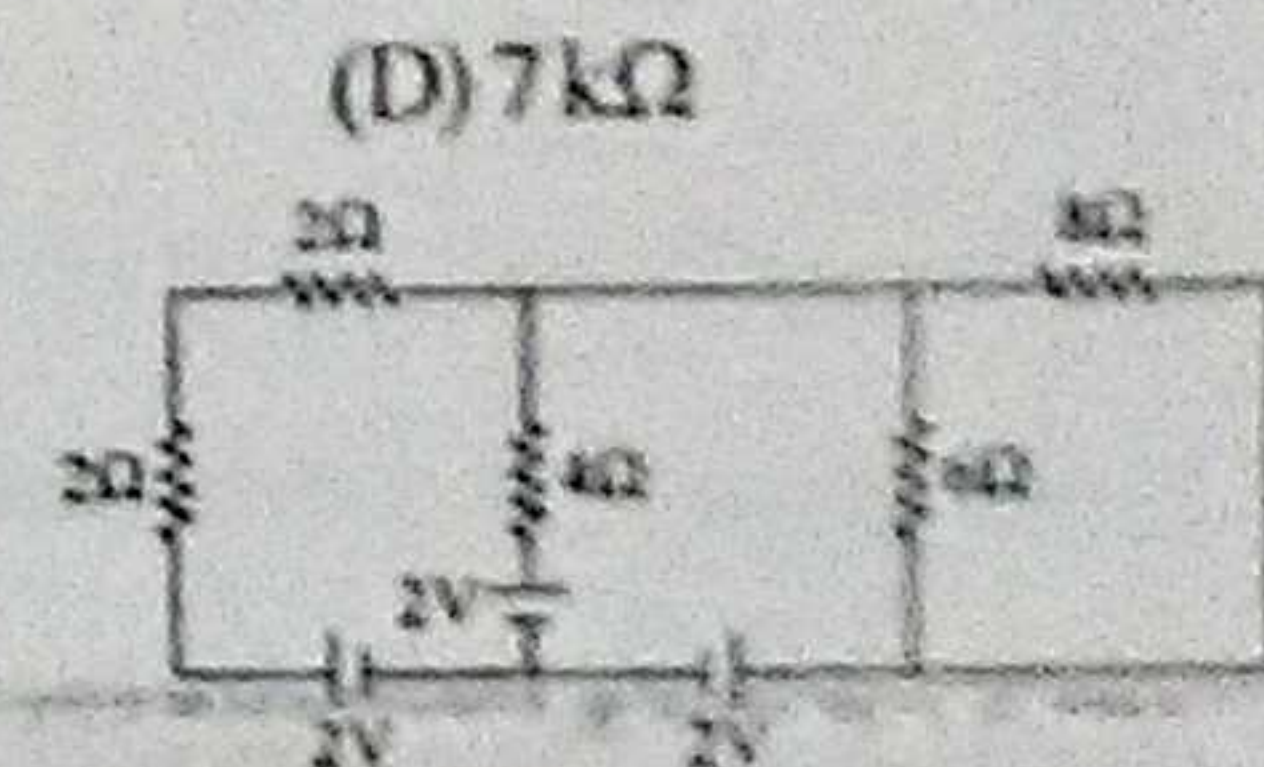
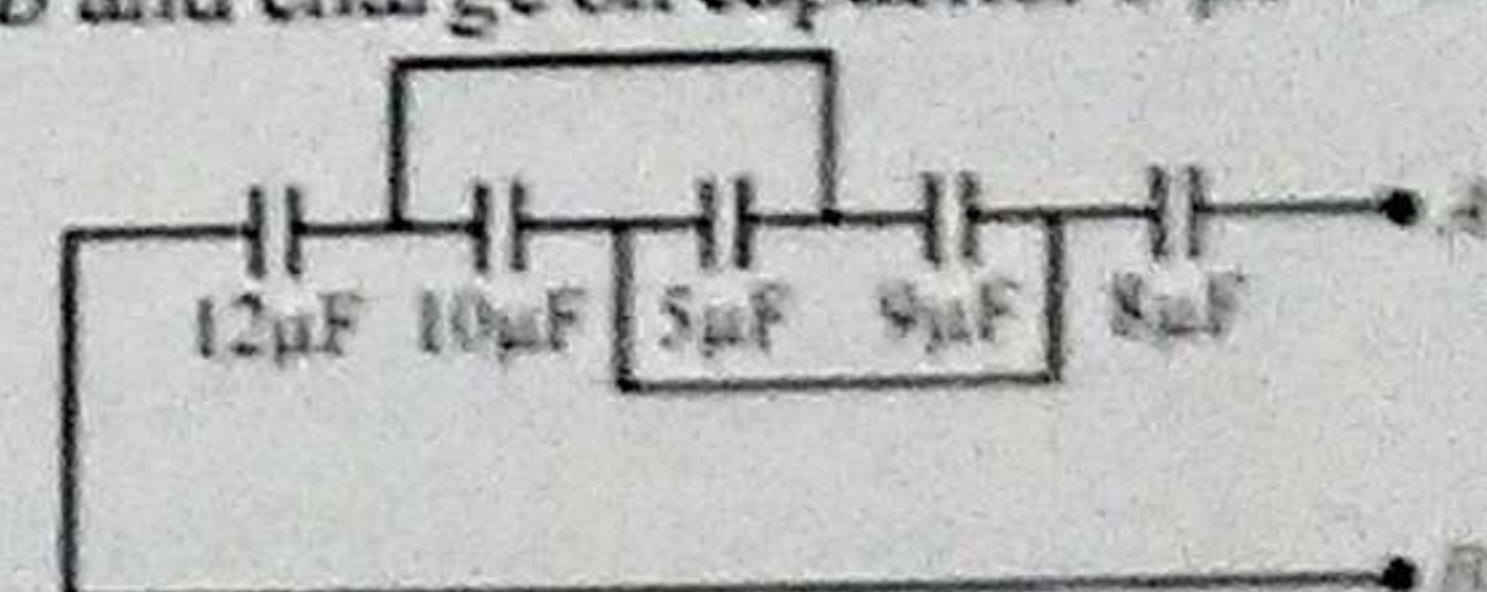
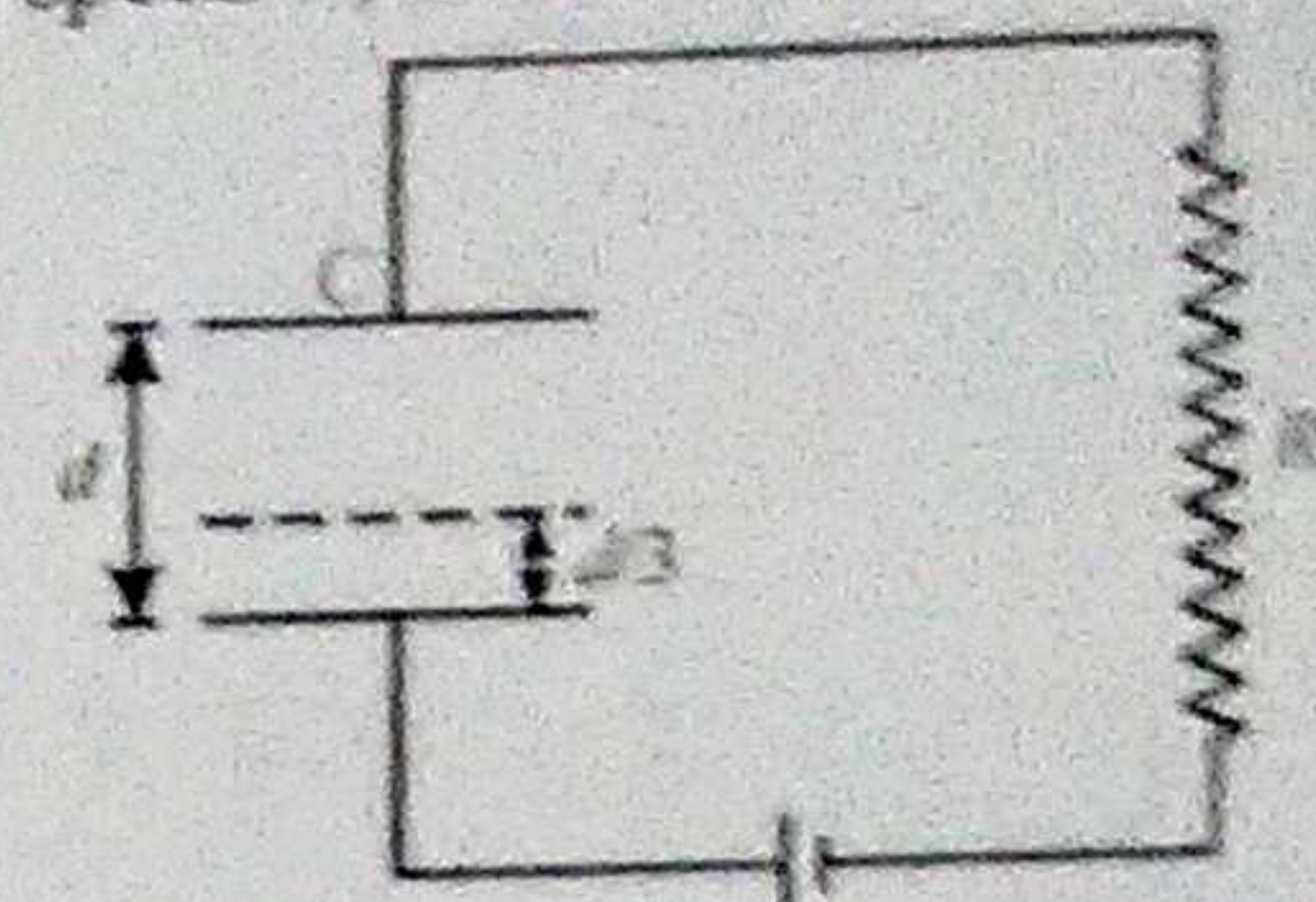


8. A circuit is connected as shown in the figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X is :

- (A) 0
(B) $54\ \mu\text{C}$
(C) $27\ \mu\text{C}$
(D) $81\ \mu\text{C}$



9. A parallel plate capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant $K = 2$. The level of liquid is $\frac{d}{3}$ initially. Suppose the liquid level decreases at a constant speed V , the time constant as a function of time t is :
- (A) $\frac{6\epsilon_0 R}{5d + 3Vt}$ (B) $\frac{(15d + 9Vt)\epsilon_0 R}{2d^2 - 3dVt - 9V^2t^2}$
 (C) $\frac{6\epsilon_0 R}{5d - 3Vt}$ (D) $\frac{(15d - 9Vt)\epsilon_0 R}{2d^2 + 3dVt - 9V^2t^2}$
10. A straight conductor of uniform cross-section carries a current I . Let $s =$ specific charge of an electron. The momentum of all the free electrons per unit length of the conductor, due to their drift velocities only, is :
 (A) Is (B) I/s (C) $\sqrt{I/s}$ (D) $(Is)^2$
11. 1 m long metallic wire is broken into two unequal parts P and Q . P of the wire is uniformly extended into another wire R . Length of R is twice the length of P and the resistance of R is equal to that of Q . Find the ratio of the resistance of P and R and also the ratio of lengths of P and Q :
12. Five capacitors together with their capacitances are shown in the adjoining figure. The potential difference between the points A and B is 60 volt. The equivalent capacitance between the point A and B and charge on capacitor $5 \mu\text{F}$ will be respectively :
 (A) $44 \mu\text{F}$, $30 \mu\text{C}$ (B) $16 \mu\text{F}$, $150 \mu\text{C}$
 (C) $15 \mu\text{F}$, $200 \mu\text{C}$ (D) $4 \mu\text{F}$, $50 \mu\text{C}$
13. A capacitor of capacitance $10 \mu\text{F}$ is connected to a battery of emf 2V . It is found that it takes 50 ms for the charge on the capacitor to become $12.6 \mu\text{C}$. Then the resistance of the circuit is :
 (A) $4 \text{ k}\Omega$ (B) $5 \text{ k}\Omega$ (C) $6 \text{ k}\Omega$ (D) $7 \text{ k}\Omega$
14. Find the currents in the different resistors shown in figure
15. The equivalent resistance between points A and B is :
 (A) $2R$ (B) $\frac{3}{4}R$
 (C) $\frac{4}{3}R$ (D) $\frac{3}{5}R$
16. In a potentiometer circuit, two wires of same material of resistivity ρ , one of radius of cross-section ' a ' other of radius of cross-section ' $2a$ ' are joined in series. They are of length l and $2l$ respectively. This combination acts as the potentiometer wire of length $3l$. The emf of the cell in the primary circuit is ϵ and internal resistance is $\frac{\rho l}{2\pi a^2}$. This cell is connected to the potentiometer wire by a conducting wire of negligible resistance with positive terminal of the cell connected to one end (call it A) of longer wire. The negative terminal of the cell is connected to one end of the smaller wire. The remaining ends of the two wires are joined together. Find :
 (i) The maximum voltage which can be balanced on the potentiometer wire.
 (ii) The balancing length, measured from point A , obtained in measurement of emf of cell of emf $\frac{\epsilon}{2}$.
 (iii) If positive terminal of cell of emf $\frac{\epsilon}{2}$ and internal resistance $\frac{\rho l}{2\pi a^2}$ is connected to point A and other terminal is joined to the junction of the two wires, then find the current through this cell.



ANSWER KEY

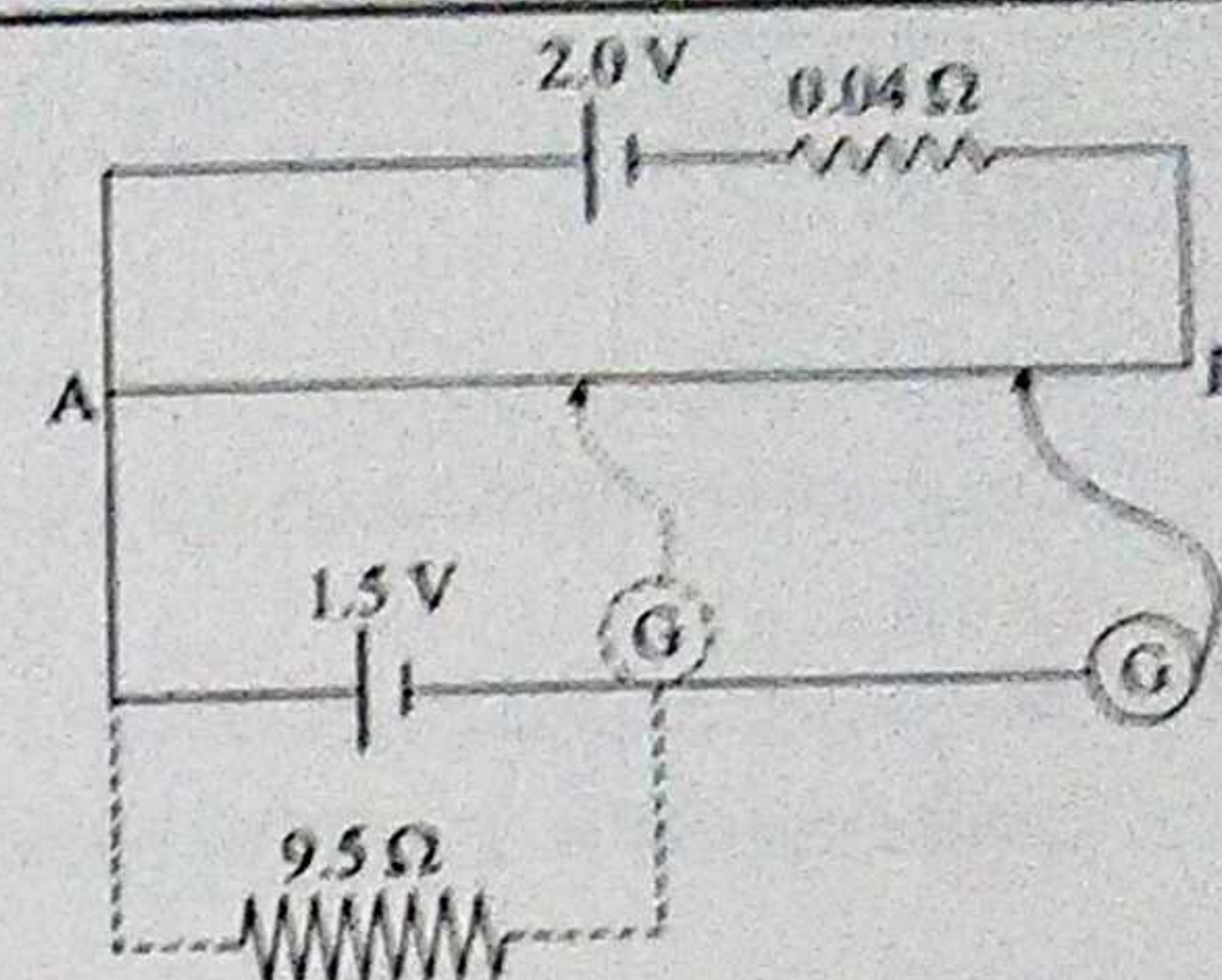
1. $[128.57 \Omega]$ 2. (C) 3. (B) 4. $[6\text{V and } -4\text{V, Zero}]$ 5. $[(a) 1\text{ A; } (b) 2/3\text{ A; } (c) 1/3\text{ A}]$ 6. (D)
 7. $[Q_0 = \frac{R_2 VC}{R_1 + R_2} \text{ and } \alpha = \frac{(R_1 + R_2)}{CR_1 R_2}]$ 8. (C) 9. (A) 10. (B) 11. $[\frac{1}{4}, \frac{1}{4}]$ 12. (D)
 13. (B) 14. [Zero] 15. (D) 16. $[(i) v_0 = \frac{3\epsilon}{4}; (ii) \frac{5l}{2}; (iii) \frac{\epsilon}{7R}, \text{ where } R = \frac{\rho l}{A} \text{ and } A = 2\pi a^2]$

CAPACITANCE & CURRENT ELECTRICITY

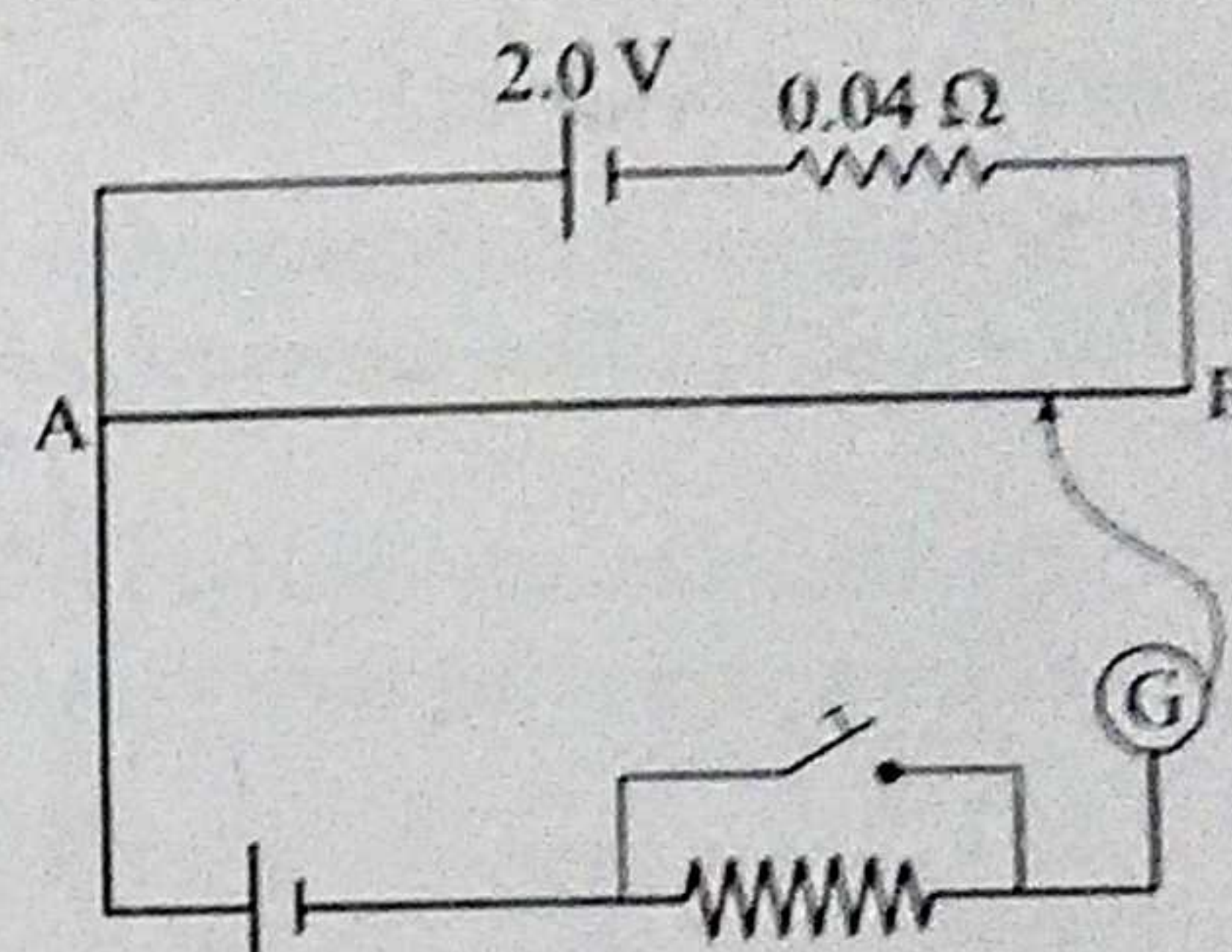
Type of CPP - Objective & Subjective

Level - MODERATE

1. Figure shows a 2.0 V potentiometer used for the determination of internal resistance of 1.5 V cell. The balance point of the cell in open circuit is 70 cm. When a resistor of 9.5Ω is used in the external circuit of the cell, the balance point shifts to 60 cm length of the potentiometer wire. Determine the internal resistance of the secondary cell.



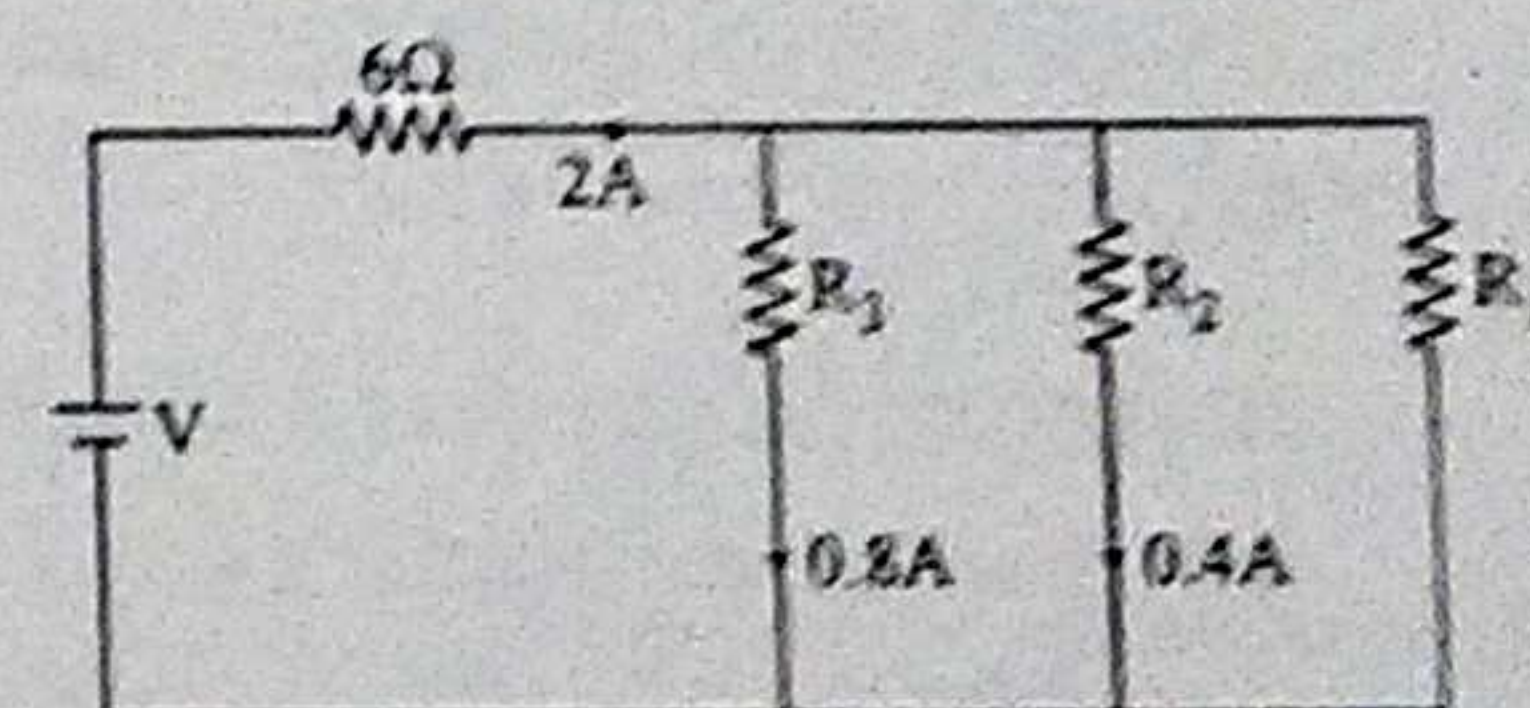
2. Figure shows a potentiometer with a cell of emf 2.0 V and internal resistance 0.04Ω maintaining a potential drop across the potentiometer wire AB. A standard cell which maintains a constant emf of 1.02 V (for very moderate currents up to a few ampere) gives a balance point of 67.3 cm length of the wire. To ensure very low currents drawn from the standard cell, a very high resistance of $600 \text{ k}\Omega$ is put in series with it which is shorted close to the balance point. The standard cell is then replaced by a cell of unknown emf E and the balance point found similarly turns out to be at 82.3 cm length of the wire :



- What is the value of E ?
- What purpose does the high resistance of $600 \text{ k}\Omega$ have?
- Is the balance point affected by this high resistance?
- Is the balance point affected by the internal resistance of the driver cell?
- Would the method work in the above situation if the driver cell of the potentiometer had an emf of 1.0 V instead of 2.0 V?
- Would the circuit work well for determining extremely small emf, say, of the order of few mV (such typical emf of thermocouple)?

3. In the circuit shown in figure :

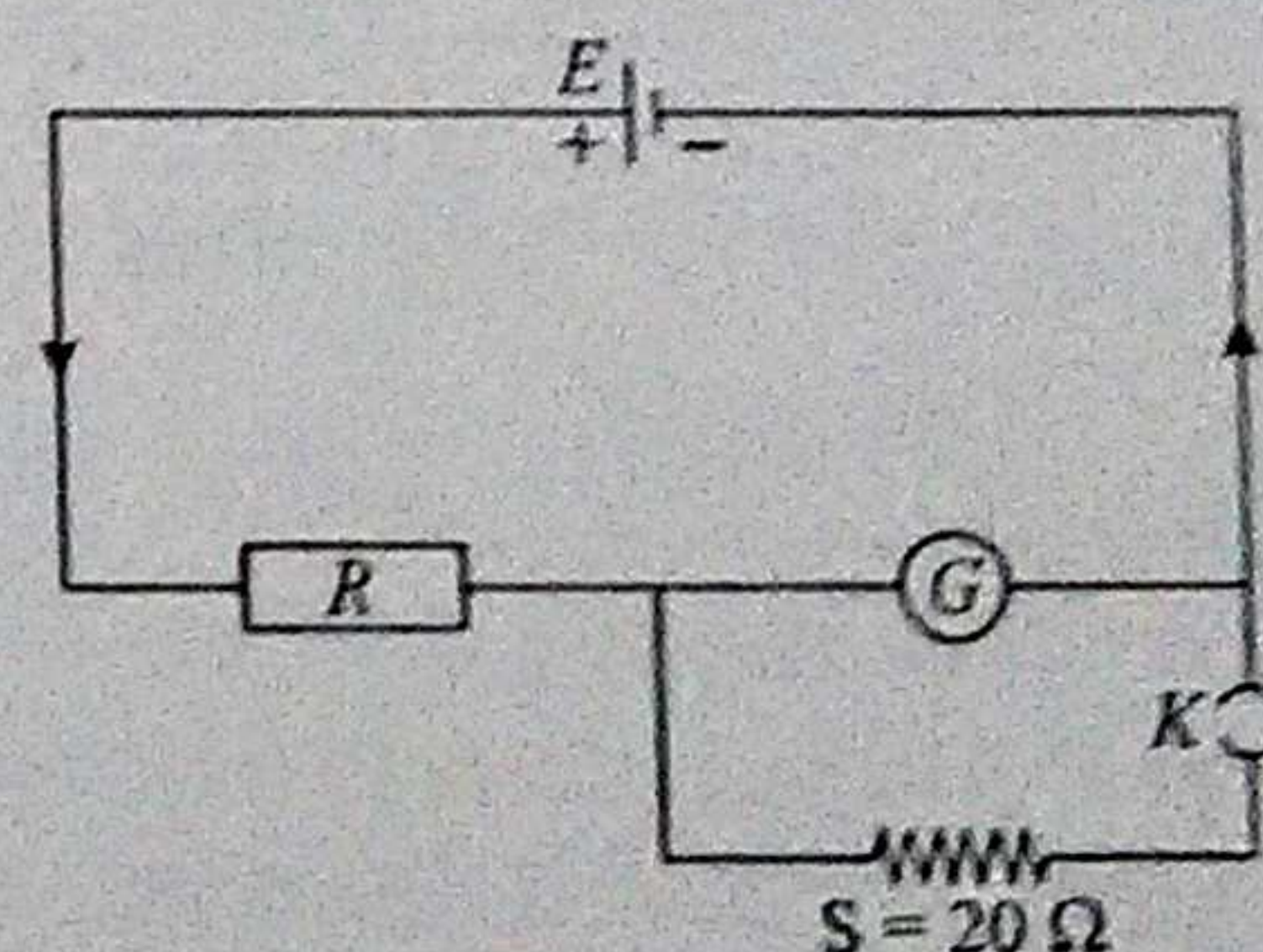
- $V = 10 \text{ V}$
- $R_1 = 10 \Omega$
- $R_2 = 20 \Omega$
- net resistance of the circuit is 10Ω



4. Suppose you have three resistors of 20Ω , 50Ω and 100Ω . What minimum and maximum resistances can you obtain from these resistors?

5. In the following circuit when we plug the key 'K' in, then the reading of the galvanometer will be half, then the resistance of the galvanometer can be :

- 20Ω
- 30Ω
- 40Ω
- 50Ω



6. In a potentiometer wire experiment the emf of a battery in the primary circuit is 20 volt and its internal resistance is 5Ω . There is a resistance box (in series with the battery and the potentiometer wire) whose resistance can be varied from 120Ω to 170Ω . Resistance of the potentiometer wire is 75Ω . The following potential differences can be measured using this potentiometer :

- 5V
- 6V
- 7V
- 8V

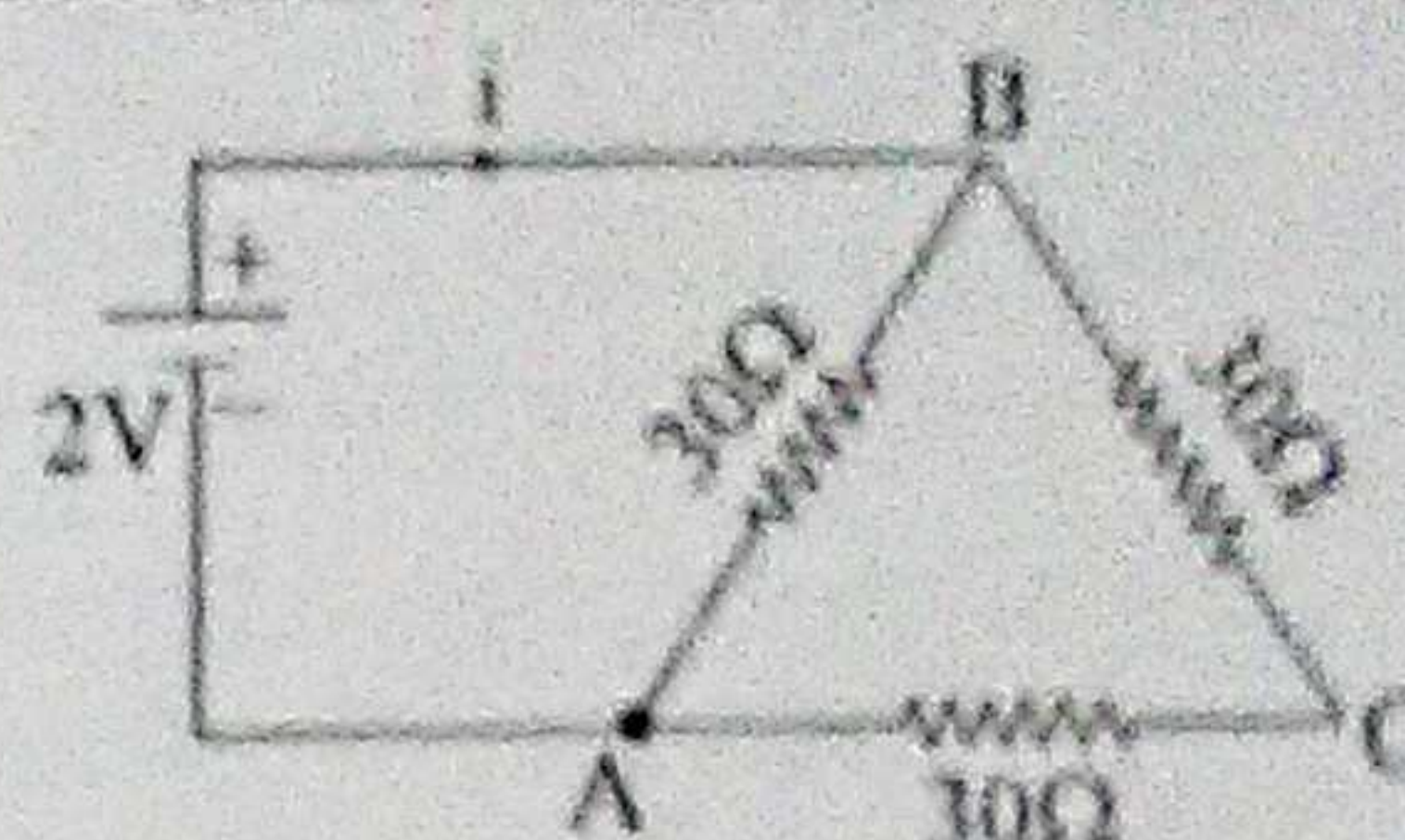
CAPACITANCE & CURRENT ELECTRICITY

Type of CPP - Objective & Subjective

Level - MODERATE

1. The current i in the circuit of fig. is :

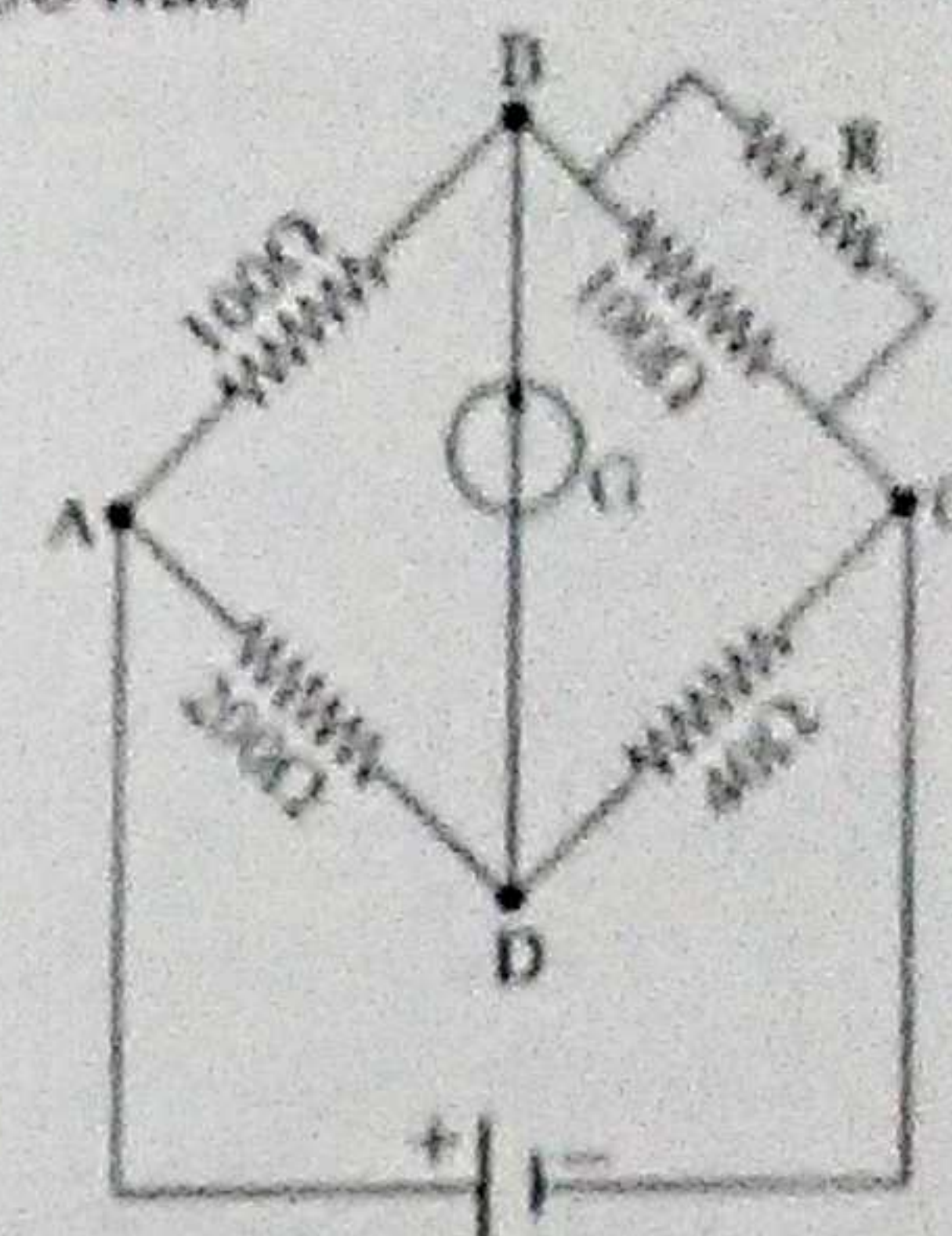
(A) $\frac{1}{45}$ amp. (B) $\frac{1}{15}$ amp.
(C) $\frac{1}{10}$ amp. (D) $\frac{1}{5}$ amp.



2. Three equal resistors connected in series across a source of emf together dissipate 10 watts of power. What would be the power dissipated if the same resistors are connected in a parallel across the same source of emf?
(A) 60 watt (B) 90 watt (C) 100 watt (D) 30 watt

3. The given Wheatstone bridge is showing no deflection in the galvanometer joined between the points B and D (Figure). Calculate the value of R :

(A) $25\ \Omega$
(B) $50\ \Omega$
(C) $40\ \Omega$
(D) $100\ \Omega$



4. (a) A car has a fresh storage battery of emf 12 V and internal resistance $5.0 \times 10^{-2}\ \Omega$. If the starter draws a current of 90 A, what is the terminal voltage of the battery when the starter is on?
(b) After long use, the internal resistance of the storage battery increases to $500\ \Omega$. What maximum current can be drawn from the battery? Assume the emf of the battery to remain unchanged.
(c) If the discharged battery is charged by an external emf source, is the terminal voltage of the battery during charging greater or less than its emf 12 V?

5. The plates of a capacitor of capacitance $10\ \mu\text{F}$, charged to $60\ \mu\text{C}$, are joined together by a wire of resistance $10\ \Omega$ at $t = 0$, then :

- (i) the charge on the capacitor in the circuit at $t = 100\ \mu\text{s}$ is :

(A) $120\ \mu\text{C}$ (B) $60\ \mu\text{C}$ (C) $30\ \mu\text{C}$ (D) $44\ \mu\text{C}$

- (ii) the charge on the capacitor in the circuit at $t = 100\ \mu\text{s}$ is :

(A) $120\ \mu\text{C}$ (B) $60\ \mu\text{C}$ (C) $22\ \mu\text{C}$ (D) $18\ \mu\text{C}$

- (iii) the charge on the capacitor in the circuit at $t = 1.0\ \text{ms}$ is :

(A) $0.003\ \mu\text{C}$ (B) $60\ \mu\text{C}$ (C) $44\ \mu\text{C}$ (D) $18\ \mu\text{C}$

6. A bulb is connected to an ideal battery of emf 10 V so that the resulting current is 10 mA. When the bulb is connected to 220 V mains (ideal), the current is 50 mA. Choose the correct alternative (s) :

(A) In the first case, the resistance of the bulb is $1\ \text{k}\Omega$ and in second case, it is $4.4\ \text{k}\Omega$.
(B) It is not possible since ohm's law is not followed
(C) the increase in resistance is due to heating of the filament of the bulb when it is connected to 220 V mains
(D) None of these

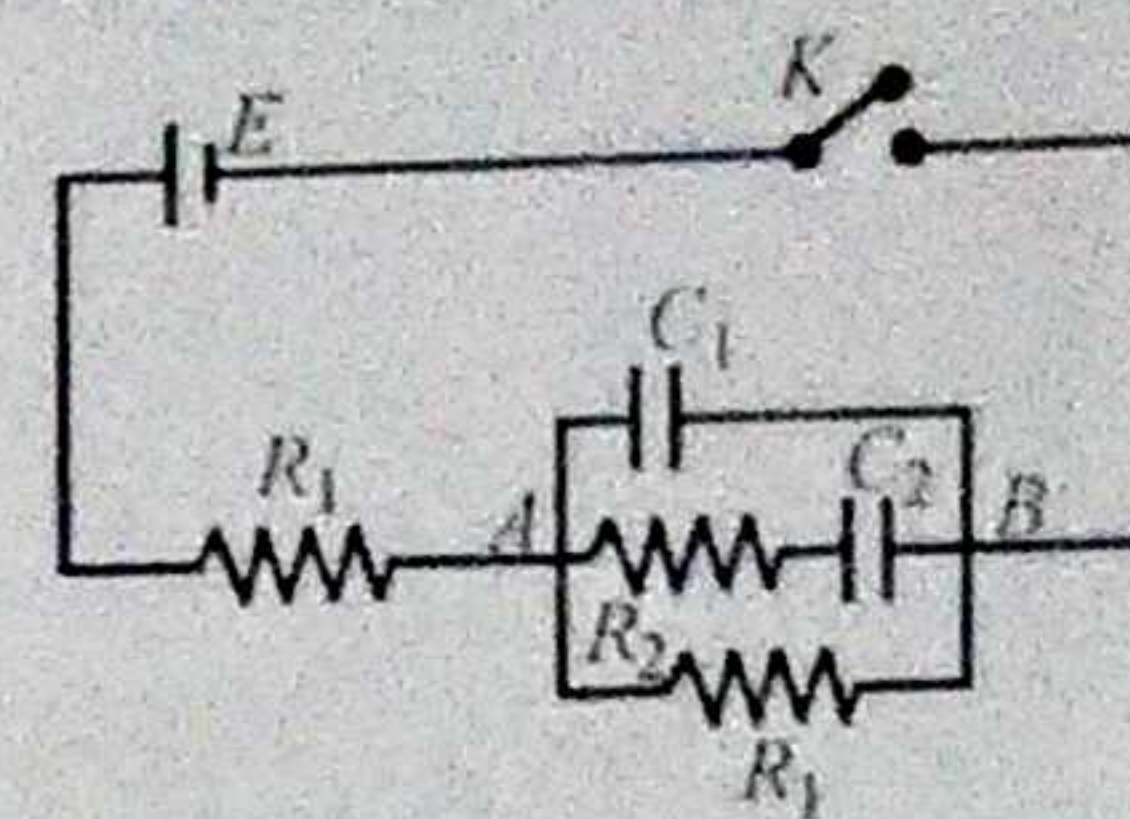
7. A network of uncharged capacitors and resistances is shown
Current through the battery immediately after key K is closed and after a long time interval is :

(A) $\frac{E}{R_1}, \frac{E}{R_1 + R_3}$

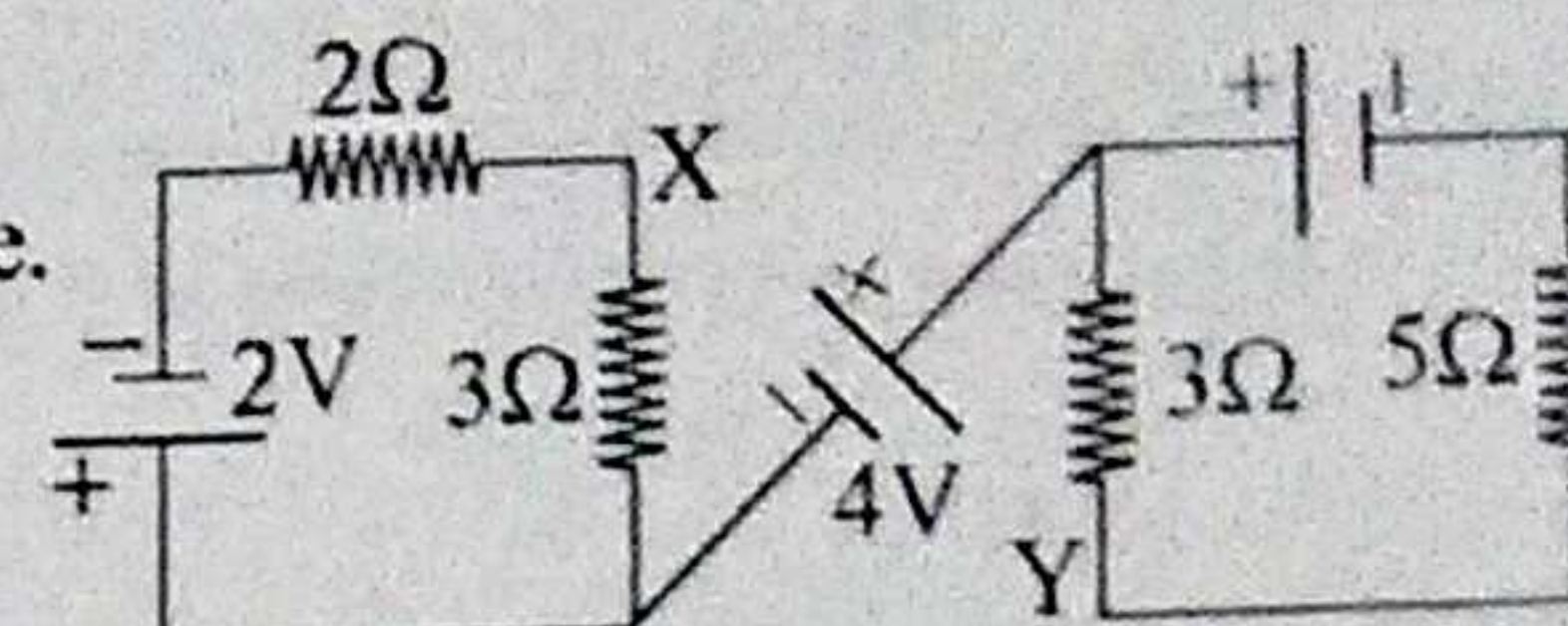
(B) $\frac{E}{R_1 + R_3}, \frac{E}{R_1 + \frac{R_2 R_3}{R_2 + R_3}}$

(C) Zero, $\frac{E}{R_1}$

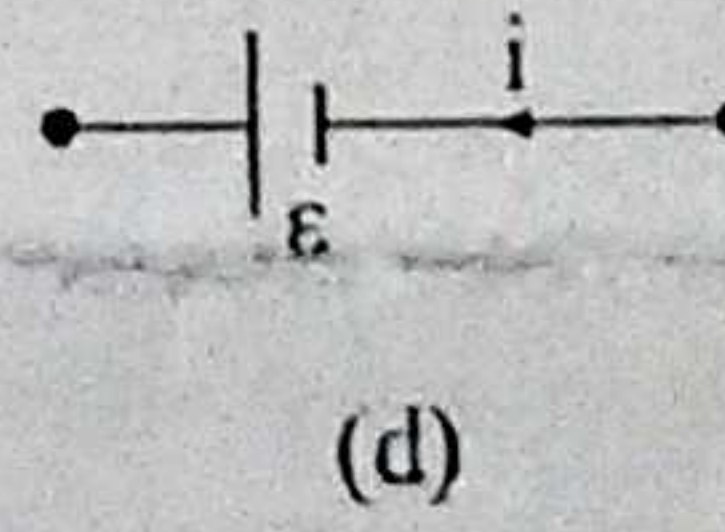
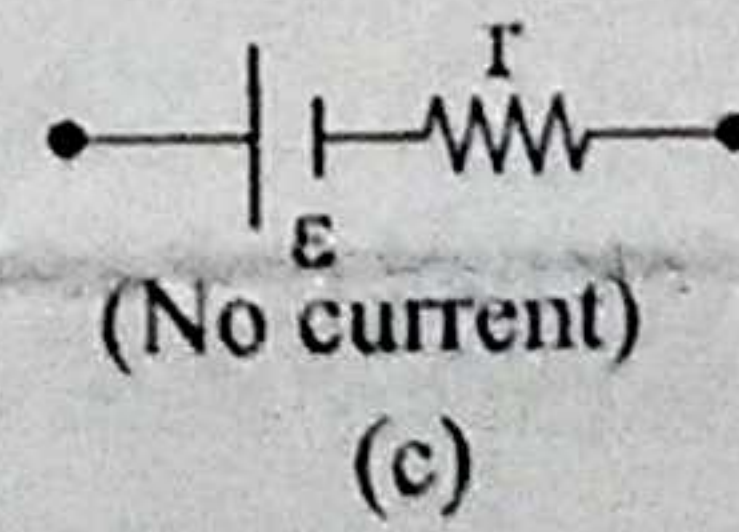
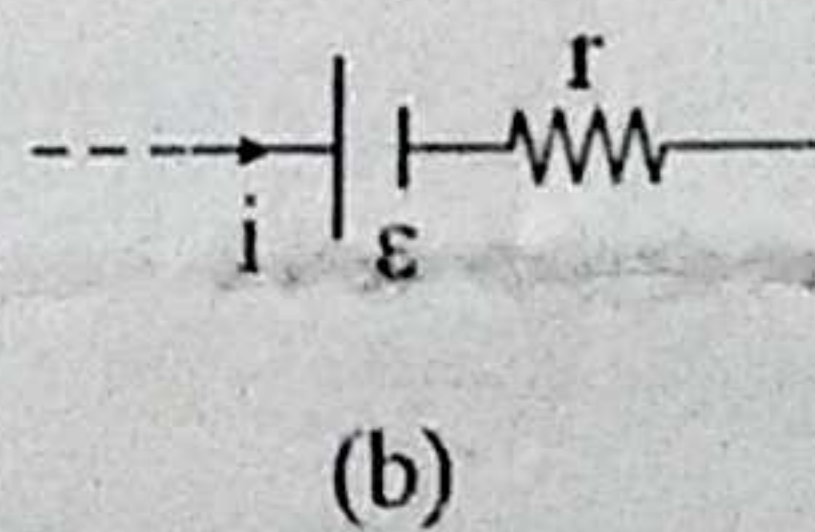
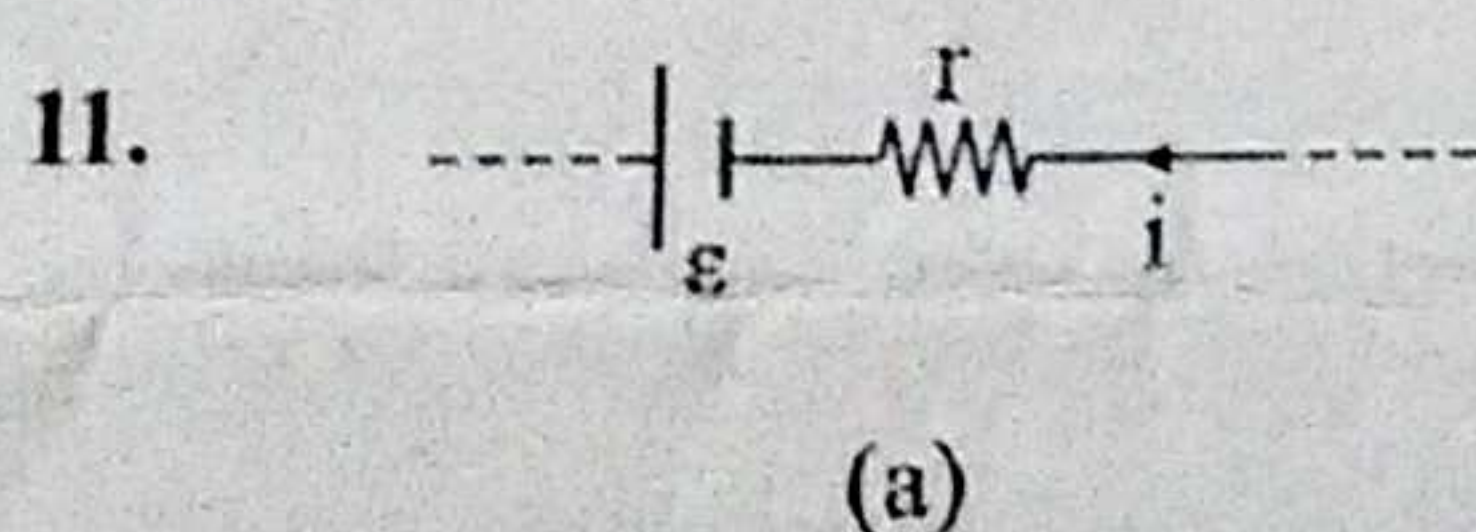
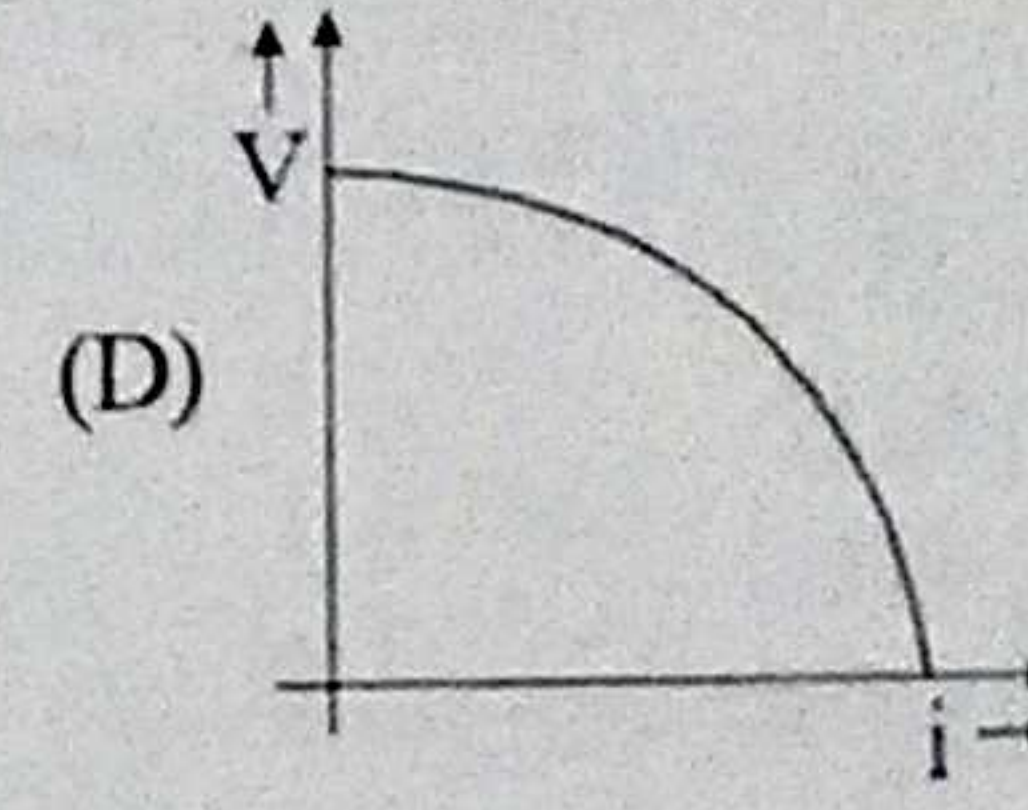
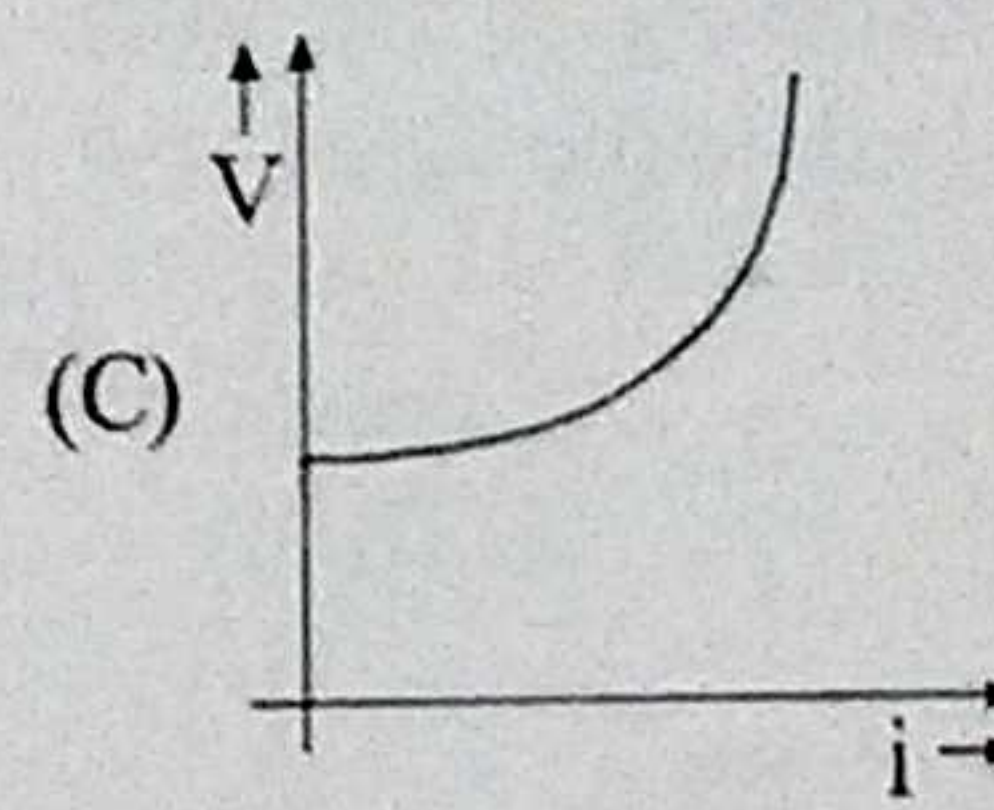
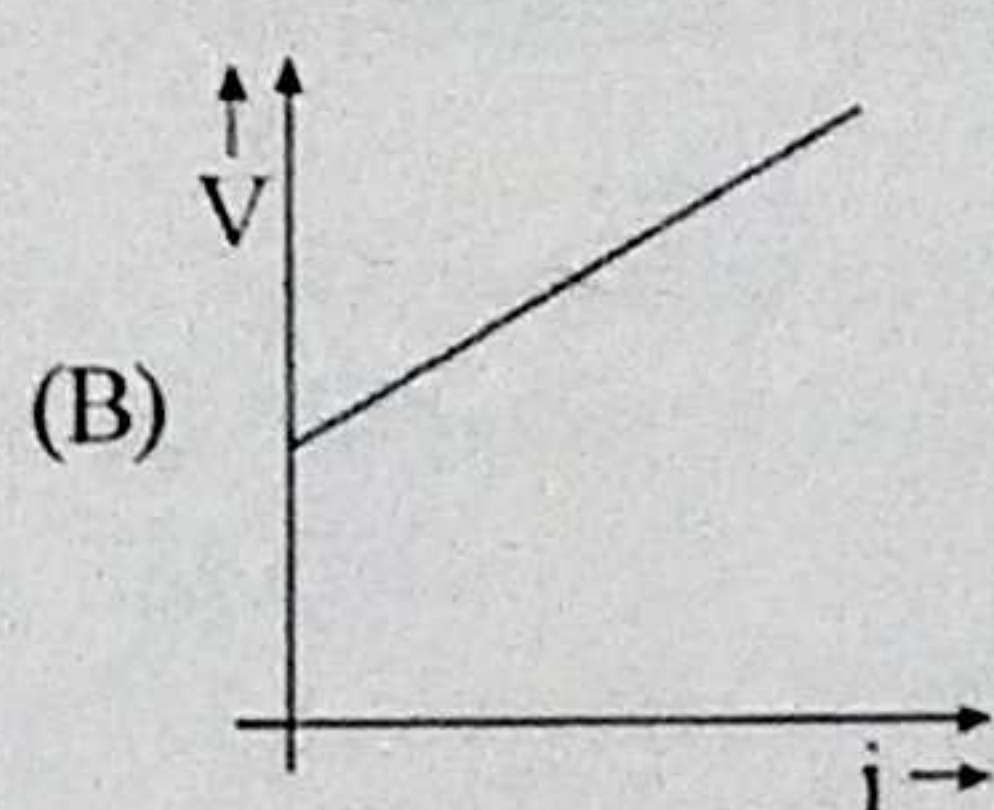
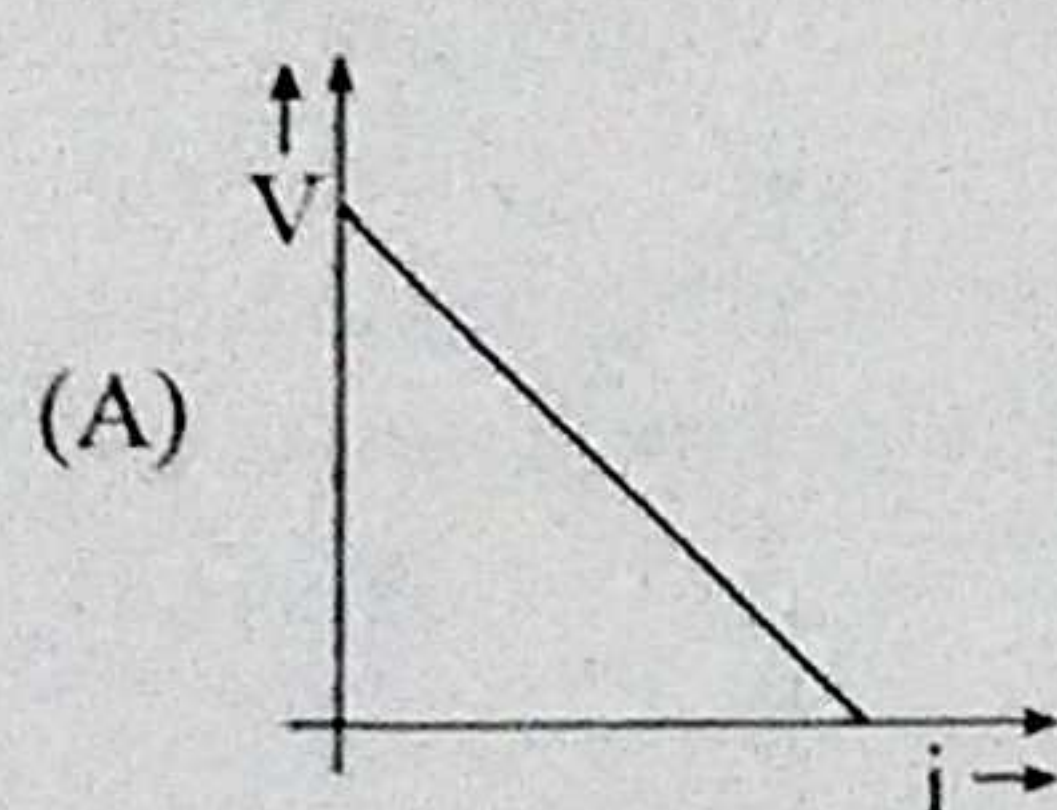
(D) $\frac{E}{R_1 + \frac{R_2 R_3}{R_2 + R_3}}, \frac{E}{R_1}$



8. (a) Determine the potential drop between X and Y in the circuit shown in Figure.
(b) If intermediate cell has internal resistance $r = 1\Omega$ then determine the potential difference between x and y .



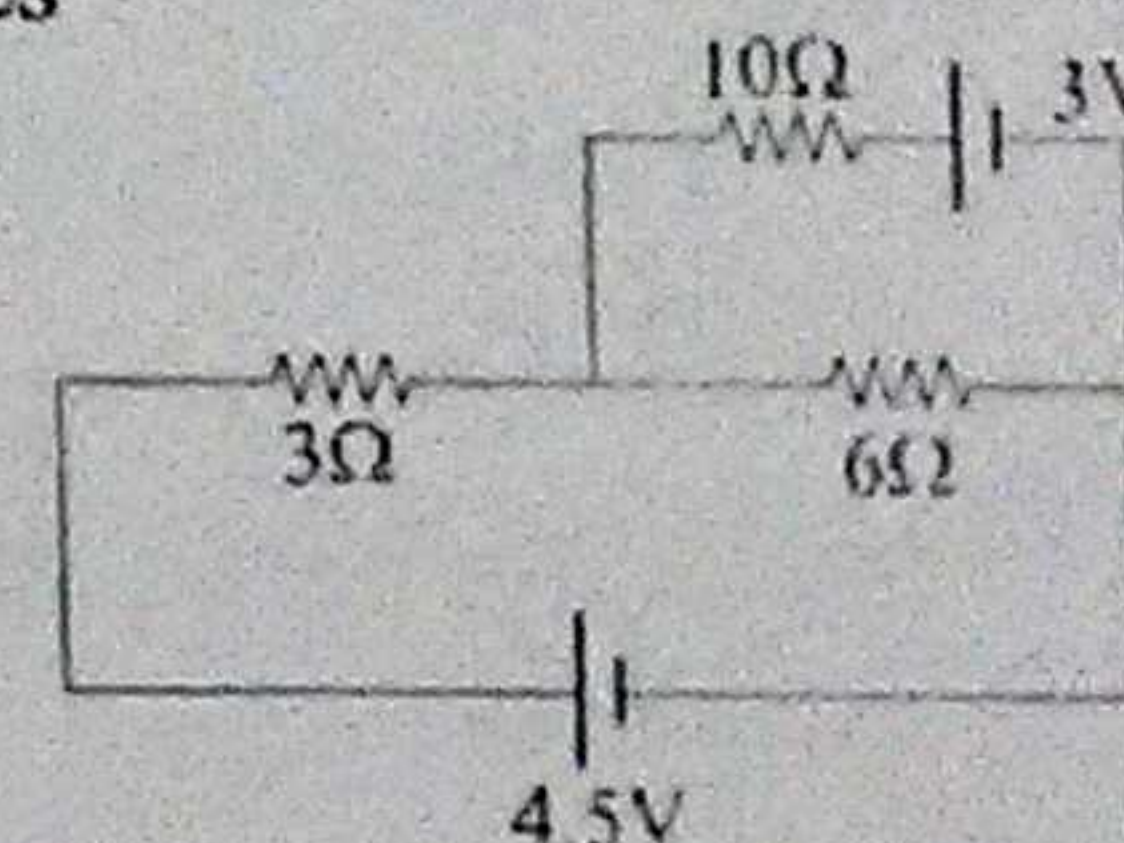
9. In an electric circuit containing a battery, the positive charge inside the battery :
(A) Always goes from the positive terminal to the negative terminal
(B) May go from the positive terminal to the negative terminal
(C) Always goes from the negative terminal to the positive terminal
(D) Does not move.
10. If internal resistance of a cell is proportional to current drawn from the cell. Then the best representation of terminal potential difference of a cell with current drawn from cell will be :



- In which of the above cells, the potential difference between the terminals of a cell exceeds its emf :
(A) a (B) b (C) c (D) d

12. A resistor of resistance R is connected to a cell of internal resistance 5Ω . The value of R is varied from 1Ω to 5Ω . The power consumed by R :
(A) Increases continuously (B) Decreases continuously
(C) First decreases then increases (D) First increases then decreases

13. Find the current through the 10Ω resistor shown in figure :
(A) Zero
(B) 1A
(C) 2A
(D) 5A



14. The efficiency of a cell when connected to a resistance R is 60%. What will be its efficiency if the external resistance is increased to six times :
(A) 80% (B) 90% (C) 55% (D) 95%
15. A wire of resistance 0.1 ohm cm^{-1} bent to form a square $ABCD$ of side 10 cm. A similar wire is connected between the corners B and D to form the diagonal BD . Find the effective resistance of this combination between corners A and C . If a 2V battery of negligible internal resistance is connected across A and C calculate the total power dissipated :
(A) $1\Omega, 3\text{ W}$ (B) $1\Omega, 4\text{ W}$ (C) $2\Omega, 4\text{ W}$ (D) $2\Omega, 4\text{ W}$

* * * * *

ANSWER KEY

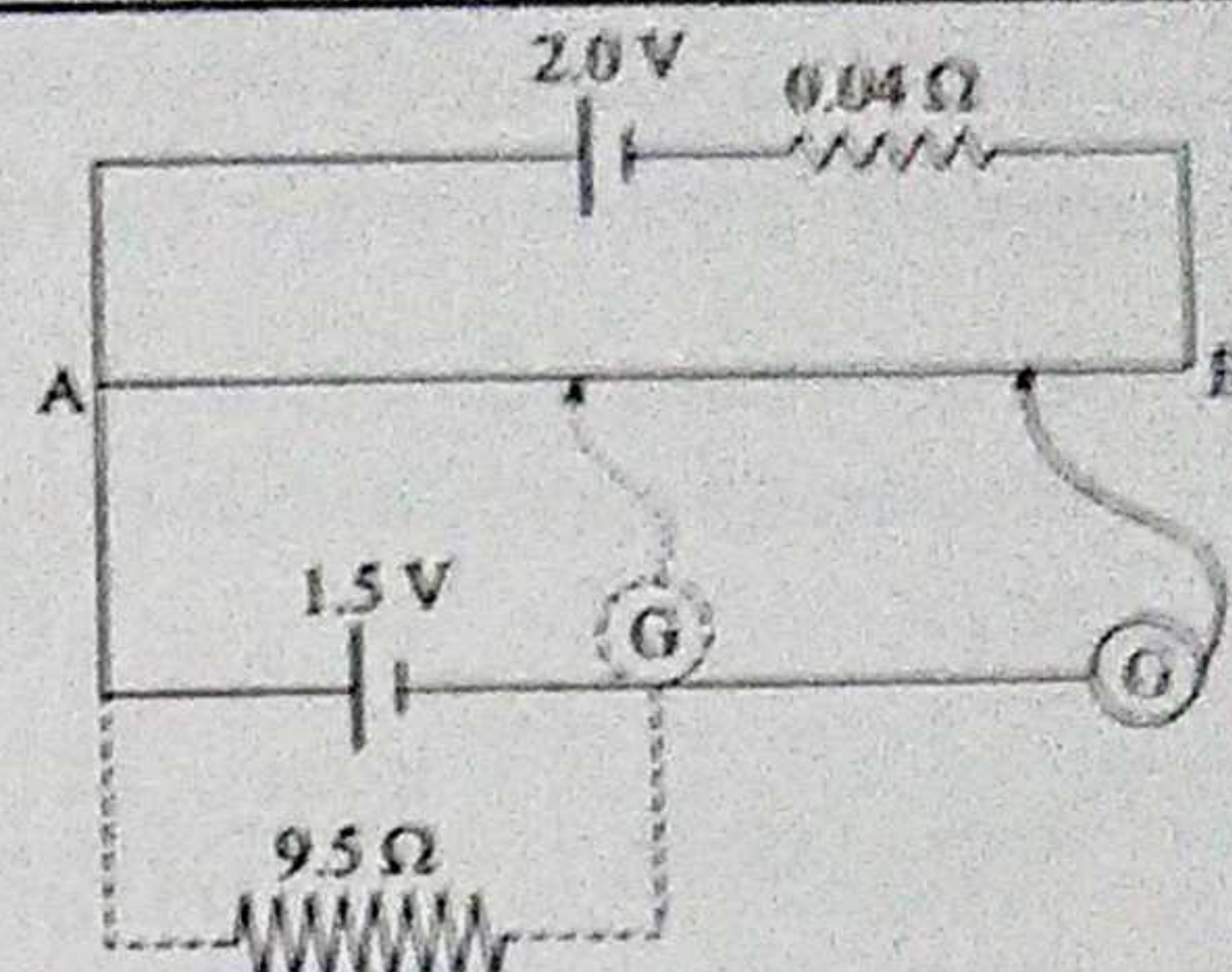
1. (C) 2. (B) 3. (A) 4. [(a) 7.5 V, (b) 24 mA; (c) greater than 12 V] 5. (i) (B) (ii) (C)
(iii) (A) 6. (A)(C) 7. (A) 8. [(a) 3.7 V; (b) 3.7 V] 9. (B) 10. (D) 11. (B)
12. (A) 13. (A) 14. (B) 15. (B)

CAPACITANCE & CURRENT ELECTRICITY

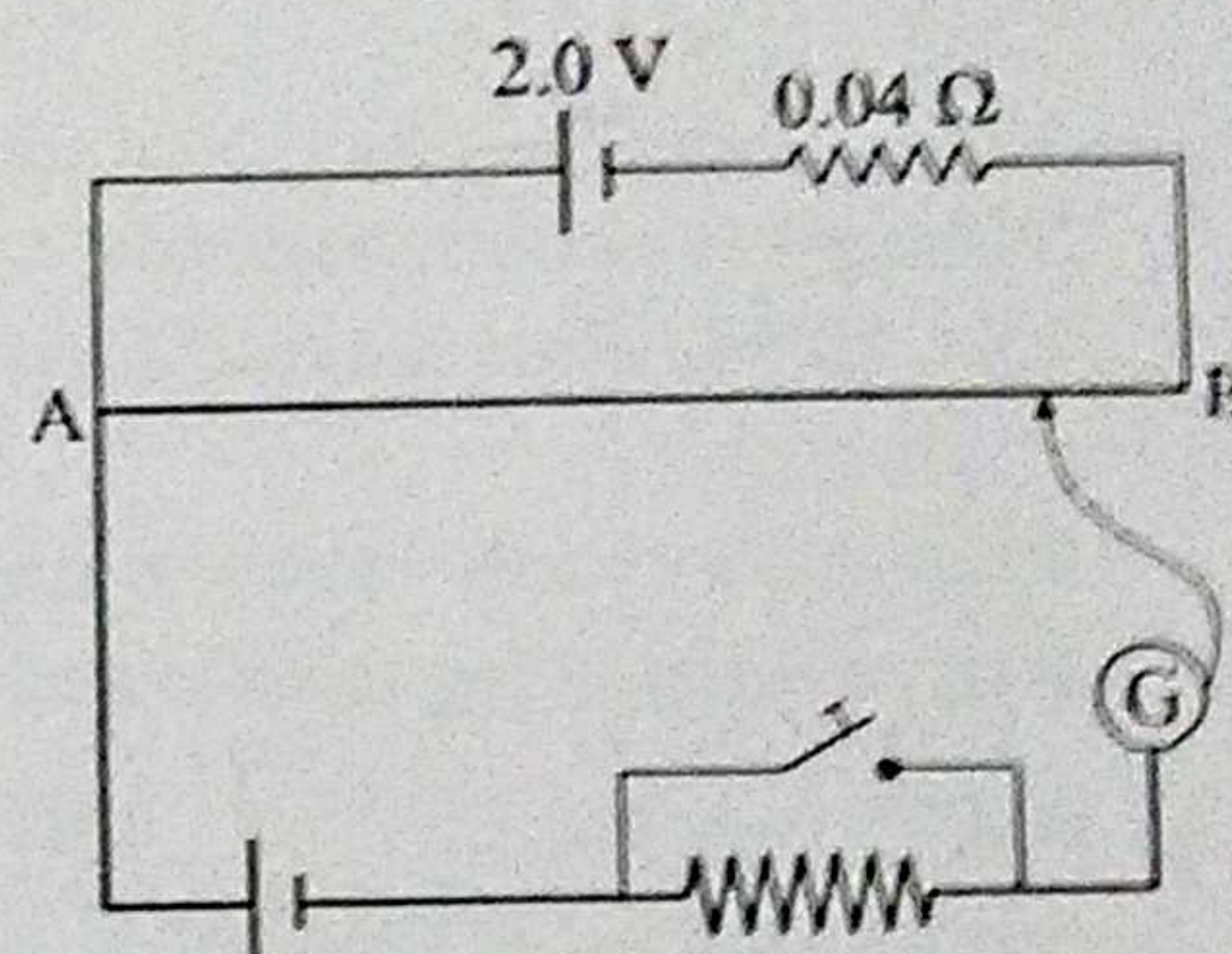
Type of CPP - Objective & Subjective

Level - MODERATE

1. Figure shows a 2.0 V potentiometer used for the determination of internal resistance of 1.5 V cell. The balance point of the cell in open circuit is 70 cm. When a resistor of 9.5Ω is used in the external circuit of the cell, the balance point shifts to 60 cm length of the potentiometer wire. Determine the internal resistance of the secondary cell.



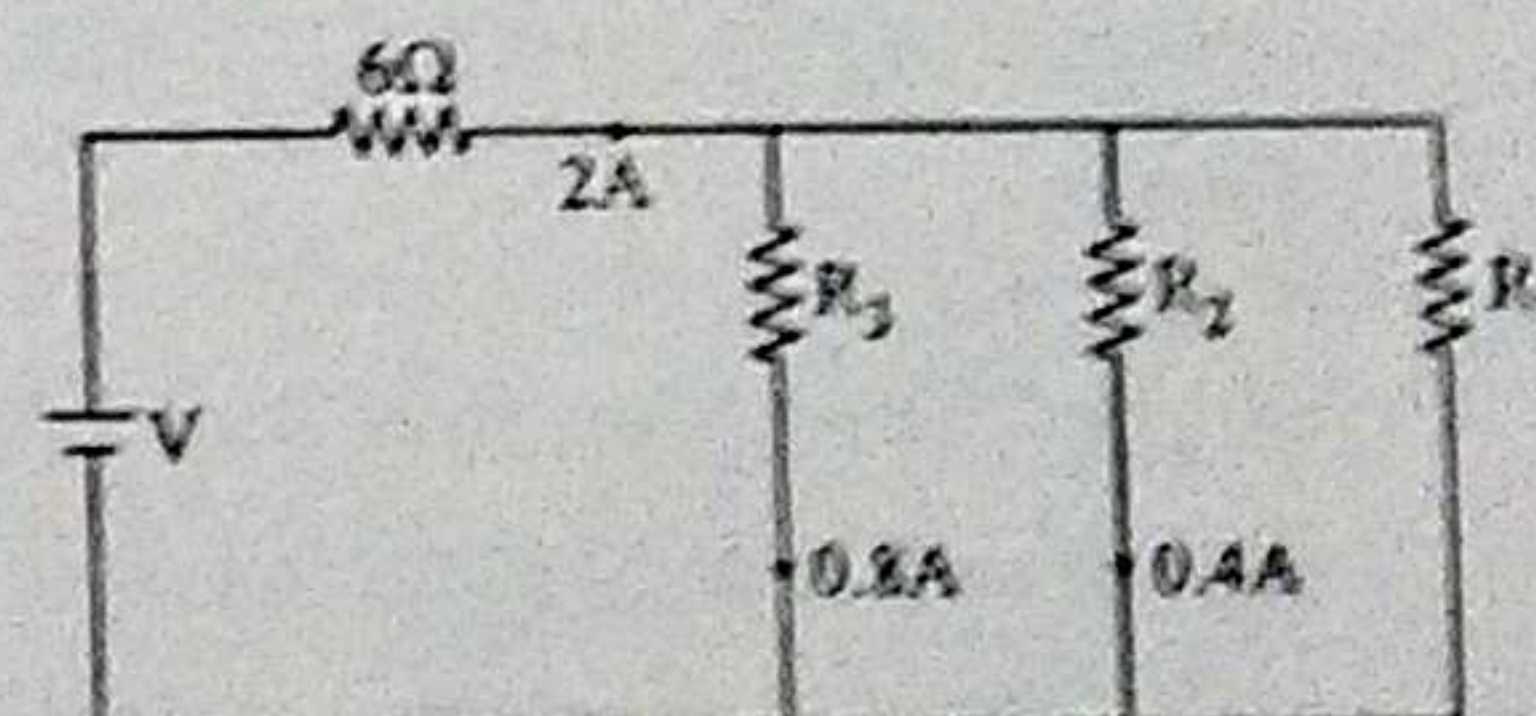
2. Figure shows a potentiometer with a cell of emf 2.0 V and internal resistance 0.04Ω maintaining a potential drop across the potentiometer wire AB. A standard cell which maintains a constant emf of 1.02 V (for very moderate currents up to a few ampere) gives a balance point of 67.3 cm length of the wire. To ensure very low currents drawn from the standard cell, a very high resistance of $600 \text{ k}\Omega$ is put in series with it which is shorted close to the balance point. The standard cell is then replaced by a cell of unknown emf E and the balance point found similarly turns out to be at 82.3 cm length of the wire :



- What is the value of E ?
- What purpose does the high resistance of $600 \text{ k}\Omega$ have ?
- Is the balance point affected by this high resistance ?
- Is the balance point affected by the internal resistance of the driver cell ?
- Would the method work in the above situation if the driver cell of the potentiometer had an emf of 1.0 V instead of 2.0 V ?
- Would the circuit work well for determining extremely small emf, say, of the order of few mV (such typical emf of thermocouple)

3. In the circuit shown in figure :

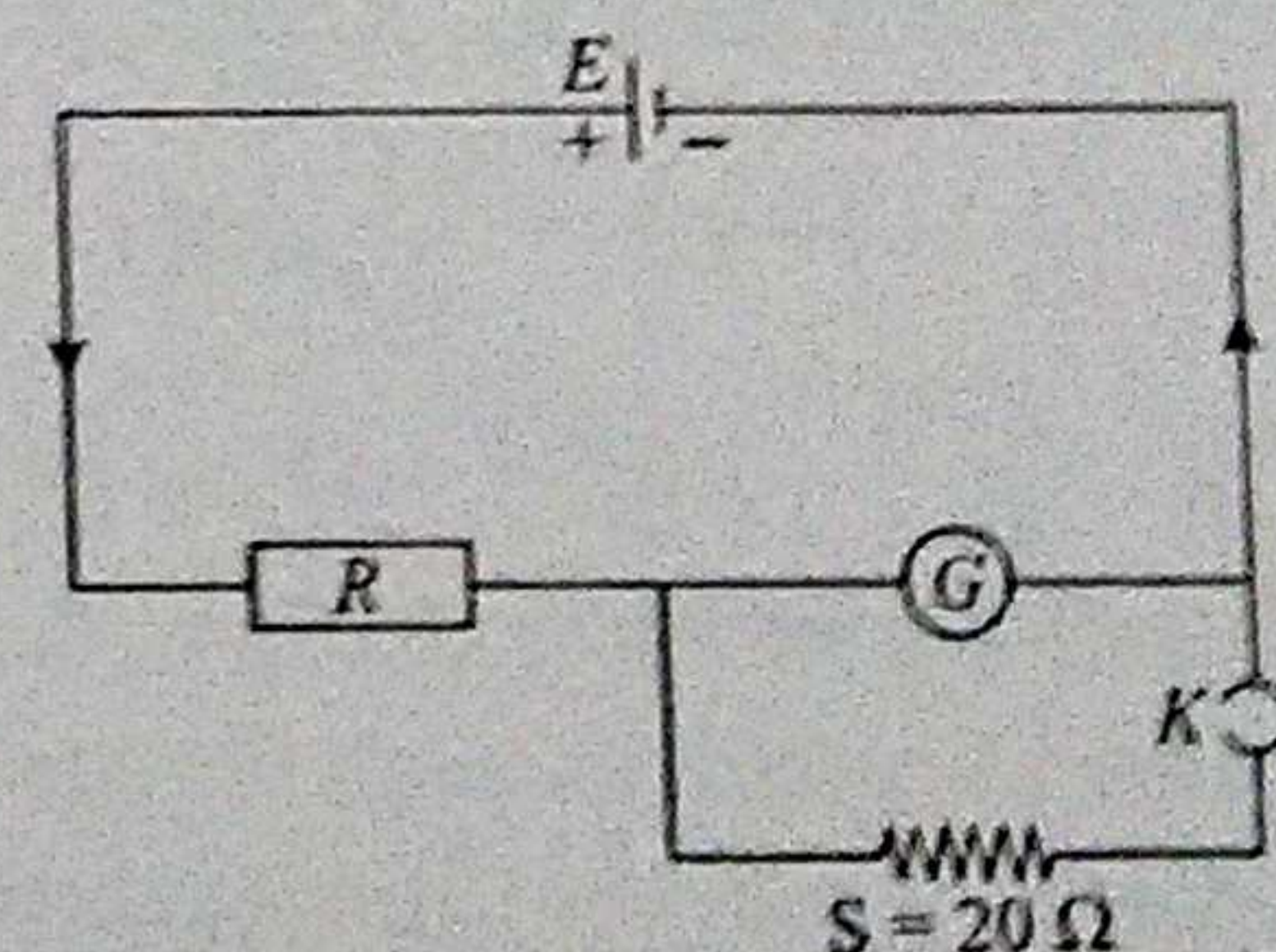
- $V = 10 \text{ V}$
- $R_1 = 10 \Omega$
- $R_2 = 20 \Omega$
- net resistance of the circuit is 10Ω



4. Suppose you have three resistors of 20Ω , 50Ω and 100Ω . What minimum and maximum resistances can you obtain from these resistors?

5. In the following circuit when we plug the key 'K' in, then the reading of the galvanometer will be half, then the resistance of the galvanometer can be :

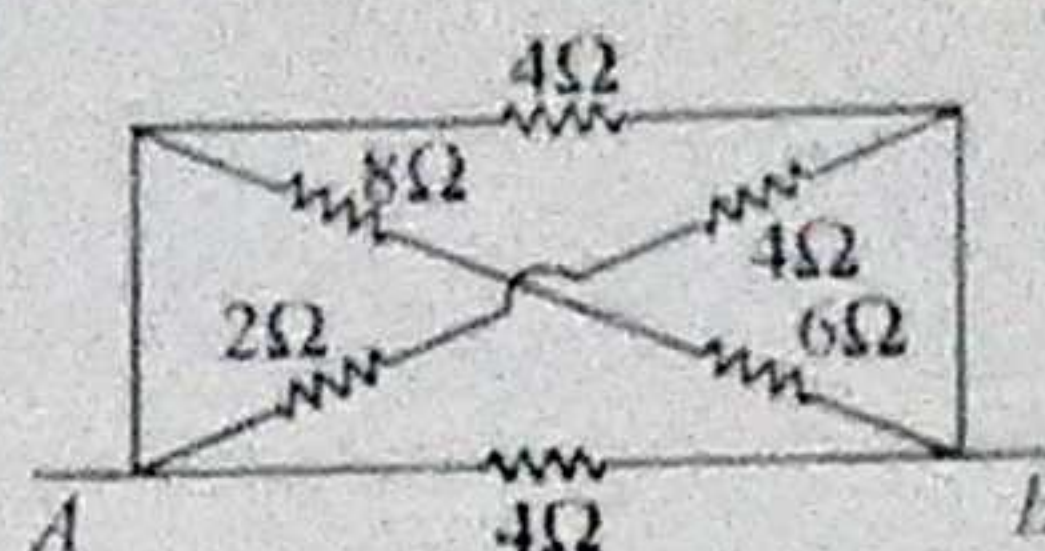
- 20Ω
- 30Ω
- 40Ω
- 50Ω



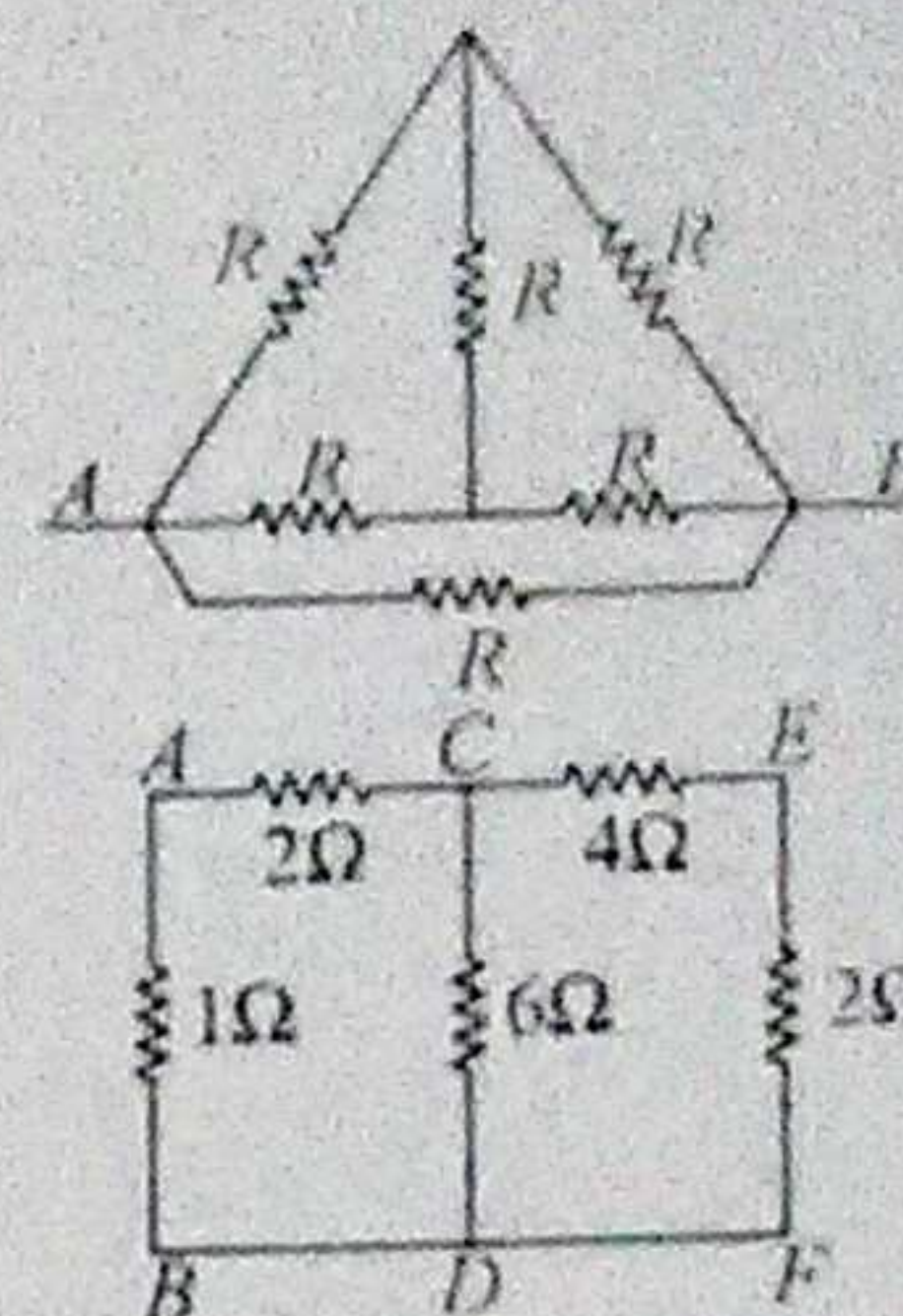
6. In a potentiometer wire experiment the emf of a battery in the primary circuit is 20 volt and its internal resistance is 5Ω . There is a resistance box (in series with the battery and the potentiometer wire) whose resistance can be varied from 120Ω to 170Ω . Resistance of the potentiometer wire is 75Ω . The following potential differences can be measured using this potentiometer :

- 5V
- 6V
- 7V
- 8V

7. Find the equivalent resistance between A and B in the following circuits.



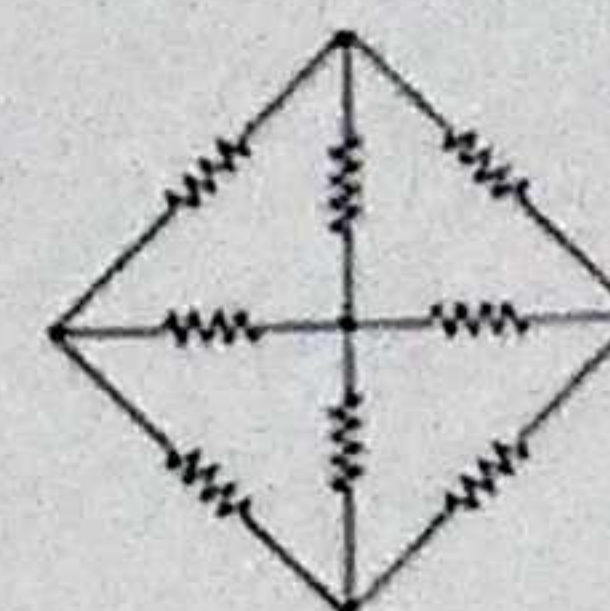
8. Find the equivalent resistance of the circuit given in figure between the following point :
(i) A & B (ii) C & D (iii) E & F (iv) A & F (v) A & C



9. A 10 m long potentiometer wire, has specific resistance S and potential gradient 20 micro-volt/cm. If the wire of $2S$ specific resistance to obtain the potential gradient of 10 micro volt/cm, the length of wire in metres should be :
(A) 50 (B) 40 (C) 25 (D) 20

10. In the network shown, each resistance is equal to R . The equivalent resistance between diagonally opposite corners is :

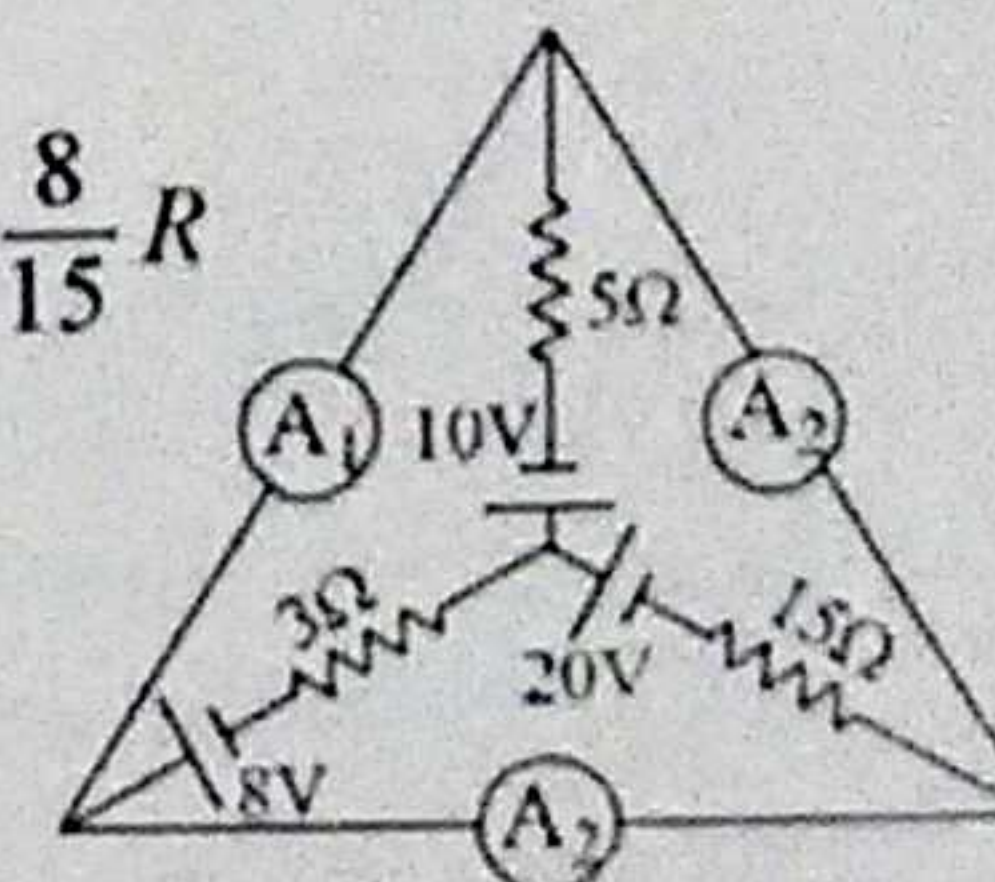
(A) R (B) $R/3$ (C) $2R/3$ (D) $4R/3$



11. In the previous question, the equivalent resistance between adjacent corners is :

(A) R (B) $\frac{2}{3}R$ (C) $\frac{3}{7}R$ (D) $\frac{8}{15}R$

12. In the given circuit the ammeter A_1 and A_2 are ideal and the ammeter A_3 has a resistance of $1.9 \times 10^{-3} \Omega$. Find the readings of all three meters.



13. There are n exactly identical resistors each having resistance R . The resultant resistance when joined in parallel is λ , then on connecting them in series the resistance will come out to be :

(A) $\frac{\lambda}{n^2}$ (B) $n^2\lambda$ (C) $\frac{\lambda}{n^3}$ (D) $n^3\lambda$

14. Three copper wires have their lengths in the ratio 5 : 3 : 1 and their masses are in the ratio 1 : 3 : 5. Their electrical resistance will be in the ratio :

(A) 5 : 3 : 1 (B) 1 : 3 : 5 (C) 125 : 15 : 1 (D) 1 : 15 : 125

15. An accumulator of 2 volts emf is connected with 10 m long potentiometer wire which has 10 ohms resistance. Under this condition, the null point for a primary cell is obtained at half the length of the potentiometer wire. Keeping the lengths of the potentiometer wire as 10 m, but its thickness is doubled, the new balancing length in metres will be :

(A) 5 (B) 2.5 (C) 7.5 (D) 1.25

16. A ring of radius R is having two charges q and $2q$ distributed on its two half parts. The electric potential at a point on its axis at a distance $2\sqrt{2}R$ from its centre is :

(A) $\frac{3kq}{R}$ (B) $\frac{kq}{3R}$ (C) $\frac{kq}{R}$ (D) $\frac{kq}{\sqrt{3}R}$

* * * * *

ANSWER KEY

1. $\left[\left(\frac{70}{60} - 1\right) \times 9.5 = \frac{9.5}{6} \text{ ohm}\right]$

2. [(a) $\frac{82.3}{67.3} \times 1.02 = 1.25 \text{ V}$; (b) The high resistance to keep the initial current low when null point is being located. This saves the standard cell from damage; (c) This high resistance does not affect the balance point because then there is no flow of current through the standard cell branch; (d) The internal resistance of driver cell affects the current through the potentiometer wire. Since potential gradient is changed, therefore, the balance point must be affected; (e) No, It is necessary that the emf of the driver cell is more than the emf of the cells; (f) This circuit will not work well for measurement of small emf. (mV) because the balanced point will be very near to end A, and percentage error in EMF measured due to length measurement would be very large $E = \frac{V}{100} l \Rightarrow \frac{dE}{E} = \frac{dl}{l}$ will be large if l is very small]

3. (B)(C)(D) 4. [12.5 Ω , 170 Ω] 5. (B)(C)(D) 6. (A)(B)(C) 7. [(a) $\frac{42}{31} \Omega$, (b) $\frac{R}{2}$]
8. [(i) $R_{AB} = 5/6 \Omega$ (ii) $R_{CD} = 1.5 \Omega$ (iii) $R_{EF} = 15 \Omega$ (iv) $R_{AF} = 5/6 \Omega$ (v) $R_{AC} = 4/3 \Omega$] 9. (D)
10. (C) 11. (D) 12. [$I_1 = 82/27$, $I_2 = 34/27$, $I_3 = 0$] 13. (B) 14. (C) 15. (A) 16. (C)